

UNIVERSITE CHEIKH ANTA DIOP

ECOLE SUPERIEURE POLYTECHNIQUE
DE DAKAR



Projet Technologie Internet

Etudiant : Alhousseynou Agne
Professeur : Monsieur Keba Gueye

Projet : Conception et Sécurisation d'un Réseau d'Entreprise Multi-AS

Avant-Propos

Les technologies IP servent à intégrer un nombre croissant d'utilisateurs, d'applications, d'équipements divers et des services. Ce sont des stratégies de déploiements qui favorisent la collaboration et l'échange de ressources à l'échelle mondiale. On entend souvent parler de la technologie IP, mais quelles sont réellement ses utilités ? Comment fonctionne-t-elle ? Qu'est-ce que la transmission par voix IP ? En fait, on utilise beaucoup cette technologie au quotidien, même peut-être plus que l'on ne le pense... C'est dans ce sens que nous allons mettre en œuvre un réseau basé sur l'utilisation d'un certains nombres de protocole

I. Description du Projet :

Ce projet vise à mettre en place une architecture réseau multi-AS avec une optimisation du routage et des services réseau. Il permet d'approfondir les connaissances en routage statique et dynamique (RIP, OSPF, EIGRP, BGP) ainsi qu'en services réseau (NAT, DHCP, DNS).

Le réseau sera divisé en cinq AS distincts (Le AS 4567 qui sera divisé en deux sous AS, Sub-AS), chacun ayant des caractéristiques et des exigences spécifiques :

AS1 - : Dans cet AS, une adresse constitue l'adresse des réseaux pour les clients locaux et l'adresse 203.0.113.0/30 est l'adresse IP publique fournie par l'ISP. Le Protocole **OSPF** sera utilisé pour cet AS

AS23- Routeurs avec EIGRP : Dans cet AS, nous configurerons une topologie de deux routeurs utilisant le protocole EIGRP pour la communication interne au sein de l'AS23.

AS4567

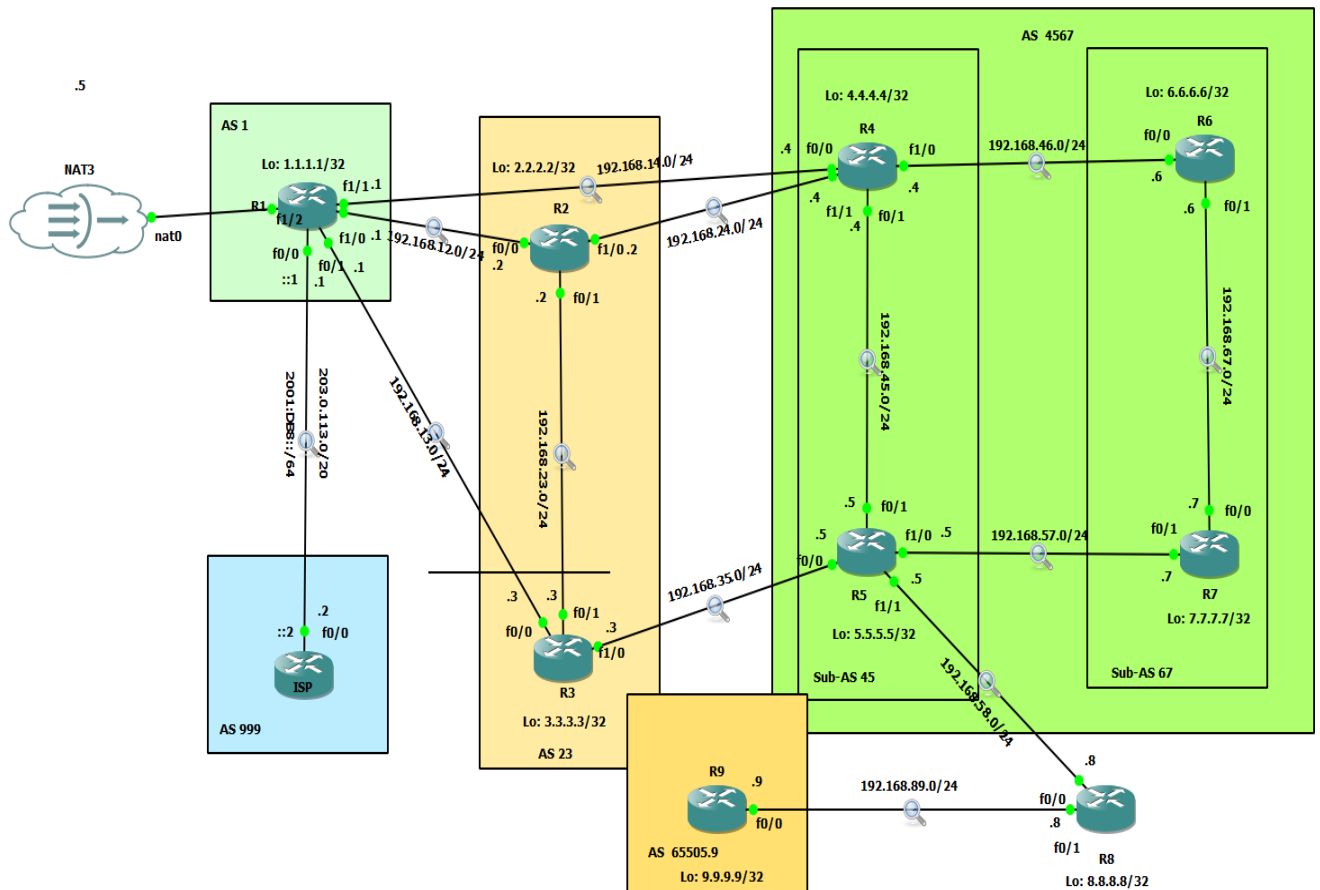
- **Sub-AS45 : Routeurs avec RIP :** Dans cet AS, nous configurerons une topologie de deux routeurs utilisant le protocole RIP pour la communication interne au sein de l'AS45.
- **Sub-AS67 : Routeurs avec OSPF :** Dans cet AS, nous configurerons une topologie de deux routeurs utilisant le protocole OSPF pour la communication interne au sein de l'AS67.

AS65505 – Routeur avec BGP : Dans cet AS, nous configurerons L'AS 65505, connecté à l'AS4567 via le routeur R8, assure l'interconnexion BGP finale avec des politiques de filtrage strictes pour sécuriser les échanges de routes et optimiser le trafic inter-domaines. C'est un Point de sortie/entrée pour le trafic externe (ex: vers d'autres opérateurs)

II) Objectifs du Projet :

1. Implémenter une architecture réseau multi-AS avec BGP pour l'interconnexion.
2. Appliquer des politiques de routage avancées (filtrage, redistribution, optimisation des routes)
3. Assurer la redondance et la gestion du trafic inter-AS.
4. Déployer des services réseau comme le NAT, DHCP et DNS.
5. Sécuriser les communications inter-AS avec des Calls et des mécanismes de protection.

Topologie proposé



PLAN de Configuration de chaque AS.....6

I. TOPOLOGIE RESEAUX

A-As 1..... 13

Configuration Routeurs.....13

Adressage IP.....13

Configuration du protocole os pf sur R1.....14

Table de routage.....	16
B-As23.....	17
Configuration routeurs.....	18
Adressage IP.....	18
Config du protocole EIGRP sur R2 et R3..	19
Redistribution des routes.....	19
Table de routage.....	23
C-As4567.....	24
SubAS45.....	24
Configuration routeurs.....	24
Adressage IP.....	24
Config du protocole RIP sur R4 et R5.....	26
Redistribution des routes.....	26
Table de routage.....	33
Sub-AS67.....	29
Configuration routeurs.....	30
Adressage IP.....	30

Config du protocole OSPF sur R6 et R7...30

Redistribution des routes.....31

Table de routage.....33

D-As65505.9.....34

Configuration routeur.....35

Adressage IP.....35

Config du protocole BGP sur R9.....35

Redistribution des routes.....35

Table de routage.....35

II. CONFIGURATION DES PROTOCOLES DE ROUTAGE

Intra-AS :

- a. **AS1** : OSPF. (R1)
- b. **AS23** : EIGRP. (R2 et R3)
- c. **Sub-AS45** : RIP. (R4 et R5)
- d. **Sub-AS67** : OSPF. (R6 et R7)
- e. **AS65505** : BGP (R9)
- f. **AS999** : ISP-Routeur Fournisseur d'accès Internet (BGP)

Inter-AS :

- g. **BGP** entre AS1 ↔ AS23 ↔ AS4567 ↔ AS65505.

h. **Authentification MD5** pour les sessions BGP.

III. Mise en place des services réseaux

- **DHCP** :
 - Sur **R1 (AS1)** pour attribuer des adresses IP aux clients du LAN
- **NAT** :
 - Sur **R1** pour permettre au LAN d'accéder à Internet via l'ISP (AS999).
- **DNS** :
 - Serveur DNS dans **AS4567** pour résoudre les noms entre AS.

IV. Sécurisation des réseaux

ACLs :

- a. Bloquer le trafic indésirable (ex: ICMP entre AS1 et AS23).
- b. Filtrer les accès depuis l'ISP (AS999).

Protection BGP :

- c. Filtrage des annonces de routes avec prefix-list et route-map.
- d. Authentification des sessions BGP.

V. Test et Validation

Connectivité :

- a. Vérifier les ping et traceroute entre tous les AS.
- b. Tester l'accès Internet depuis le LAN d'AS1.

Redondance :

- c. Simuler une panne (ex: couper un lien entre AS23 et AS4567).
- d. Vérifier la récupération via show IP bgp summary ou show ip route.

Notion important pour réaliser le projet

Adressage Ip

La syntaxe pour configurer une adresse IP sur un routeur varie en fonction du système d'exploitation du routeur. Pour les routeurs Cisco fonctionnant sous IOS, qui est l'un des systèmes d'exploitation de réseau les plus courants, voici un exemple de la syntaxe utilisée pour configurer une adresse IP sur une interface du routeur :

Router> enable

Router# configure terminal

Router (config)# interface [type de l'interface] [numéro de l'interface]

Router (config-if)# ip adresse [adresse IP] [masque de sous-réseau]

Router (config-if)# no shutdown

Voici une explication de chaque étape :

Router> enable : Passez en mode privilégié. Cela vous donne accès à plus de commandes.

Router# configure terminal : Entre dans le mode de configuration global, où vous pouvez apporter des modifications à la configuration du routeur.

Router (config)# interface [type de l'interface] [numéro de l'interface] : Sélectionnez l'interface que vous voulez configurer. Par exemple, interface GigabitEthernet0/0 pour la première interface Gigabit Ethernet.

Router (config-if)# IP adresse [adresse IP] [masque de sous-réseau] : Attribuez une adresse IP et un masque de sous-réseau à l'interface. Par exemple, ip address 192.168.1.1 255.255.255.0.

Router (config-if)# no shutdown : Active l'interface

Le Protocole Rip

Le protocole RIP (Routing Information Protocol) est un protocole de routage dynamique utilisé dans les réseaux locaux et étendus. Il utilise l'algorithme de routage à vecteur de distance pour déterminer le chemin le plus court entre chaque routeur en comptant le nombre de sauts (hops) jusqu'à la destination. RIP est considéré comme un protocole de routage intérieur (IGP, Interior Gateway Protocol), ce qui signifie qu'il est conçu pour opérer à l'intérieur d'un seul système autonome.

Router# configure terminal

Router (config)# router rip

Router (config-router)# version 2

Router (config-router)# network [adresse réseau]

Router (config-router)# network [adresse réseau]

Router (config-router)# exit

Pour redistribuer des routes apprises par le protocole RIP dans BGP (Border Gateway Protocol) sur un routeur :

```
Router> enable
```

```
Router# configure terminal
```

```
Router (config)# router bgp AS_NUMBER
```

```
Router (config-router)# redistribute rip metric  
METRIC_VALUE
```

le Protocole ospf

Le protocole OSPF (Open Shortest Path First) est un protocole de routage utilisé dans les réseaux informatiques pour distribuer les informations de routage entre les routeurs. Utilisant l'algorithme de Dijkstra pour calculer le chemin le plus court, OSPF est classé comme un protocole de routage à état de lien (link-state protocol), ce qui signifie qu'il maintient une base de données complète de la topologie réseau pour calculer les meilleurs chemins.

Pour configurer ospf dans un routeur

```
router> enable
```

```
Router# configure terminal
```

```
Router (config)# router ospf 1
```

```
Router (config-router)# network [adresse réseau] [masque wildcard] area 0
```

```
Router (config-router)# network [adresse réseau] [masque wildcard] area 0
```

Le `network [adresse réseau] [masque wildcard] area 0` permet d'annoncer le réseau dans la zone ospf 0.

Pour redistribuer ospf dans bgp

```
Router (config)# router bgp no_As
```

```
Router (config-router)# redistribute ospf 1 match internal external
```

```
Router (config-router)# exit
```

L'option **match internal external** spécifie que vous souhaitez redistribuer à la fois les routes internes OSPF (routes générées à l'intérieur de l'AS OSPF), les routes externes (routes OSPF redistribuées d'autres sources).

Le protocole EIGRP

EIGRP (Enhanced Interior Gateway Routing Protocol) est un protocole de routage avancé conçu par Cisco Systems. Utilisé principalement dans les réseaux d'entreprise, EIGRP est un protocole de routage dynamique qui combine les avantages des protocoles à vecteur de distance et à état de lien, offrant ainsi rapidité, fiabilité et facilité de configuration. EIGRP optimise le processus de routage en utilisant l'algorithme Diffusing Update Algorithm (DUAL) pour garantir des chemins sans boucle et une convergence rapide en cas de changement dans la topologie du réseau.

Pour configurer eigrp dans un routeur

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# router eigrp no_AS
```

```
Router(config-router)# network[adresse_reseau] [masque wildcard]
```

```
Router(config-router)# network [adresse_reseau] [masque wildcard]
```

Pour redistribuer eigrp dans bgp

```
router(config)# router bgp no_AS
```

```
Router(config-router)# redistribute eigrp AS
```

Le protocole BGP

BGP (Border Gateway Protocol) est le protocole de routage standard utilisé pour échanger des informations de routage et atteindre l'interconnectivité sur Internet. C'est un protocole de routage extérieur (Exterior Gateway Protocol, EGP), conçu pour permettre à des réseaux IP distincts (systèmes autonomes ou AS) de communiquer entre eux.

Pour configurer BGP dans un routeur ici on n'a pas besoin d'annonces les reseau on suit la syntaxe suivant

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# router bgp no_AS
```

```
Router(config-router)# neighbor adresse_ip_voisin remote-as As_du_voisin
```

Pour redistribuer bgp dans rip

```
Router(config)# router rip
```

```
Router(config-router)#version 2
```

```
Router(config-router)# redistribute bgp no_AS metric_value
```

Pour redistribuer bgp dans ospf 1

```
Router(config)# router ospf 1
```

```
Router(config-router)# redistribute bgp no_AS subnets
```

Pour redistribuer bgp dans eigrp

```
Router(config)# router eigrp 100
```

```
Router(config-router)# redistribute bgp no_AS metric [bande_passante][delai]  
[fiabilité] [ charge] [ MTU]
```

Il faut s'assurer de remplacer ces paramètre par des valeurs.

Ces paramètres correspondent à la bande passante (en kilobits par seconde), au délai (en microsecondes), à la fiabilité (sur une échelle de 1 à 255), à la charge (sur une échelle de 1 à 255), et à la MTU (en octets), respectivement.

I. TOPOLOGIE RESEAUX ET CONFIGURATION DES PROTOCOLS DE ROUTAGE

configuration de chaque As

A-As1

Objectif

Dans cette AS comme dit précédemment on va utiliser le protocole ospf au sein de l'AS et le protocole bgp pour assurer la connectivité avec l'AS999 , l'AS23 et L'AS4567 , faire la redistribution des routes et afficher la table de routage de chaque routeurs de l'AS.

A- Configurations routeurs

R1

Adressage ip

*Vers ISP

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int fa0/0
R1(config-if)#ip add 203.0.113.1 255.255.255.252
R1(config-if)#no sh
R1(config-if)#exie
*Mar  1 00:04:11.955: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar  1 00:04:12.955: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R1(config-if)#exit
R1(config)#
```

*Vers R3

```
R1(config)#int fa0/1
R1(config-if)#ip add 192.168.13.1 255.255.255.0
R1(config-if)#no sh
R1(config-if)#exit
*Mar  1 00:06:29.367: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar  1 00:06:30.367: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, chang
ed state to up
R1(config-if)#exit
```

Vers R2

```
R1(config)#int fal/0
R1(config-if)#ip add 192.168.12.1 255.255.255.0
% IP addresses may not be configured on L2 links.
R1(config-if)#no switchport
R1(config-if)#ip add 192.168.12.1 255.255.255.0
*Mar  1 00:09:10.455: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, chang
ed state to up
R1(config-if)#ip add 192.168.12.1 255.255.255.0
R1(config-if)#exit
R1(config)#
```

Vert R4

```
R1(config)#int fa1/1
R1(config-if)#ip add 192.168.14.1 255.255.255.0

% IP addresses may not be configured on L2 links.

R1(config-if)#no switchport
R1(config-if)#ip add 192.168.14.1 255.255.255.0
R1(config-if)#
*Mar  1 00:11:33.279: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/1, chang
ed state to up
R1(config-if)#no sh
R1(config-if)#exit
R1(config)#
```

*Configuration protocole ospf

```
R1(config)#router ospf 1
R1(config-router)#network 203.0.113.0 0.0.0.255 area 0
R1(config-router)#exit
R1(config)#
```

*Configurons le protocole bgp

```
R1(config)#router bgp 1
R1(config-router)#neighbor 203.0.113.2 remote-as 999
R1(config-router)#neighbor 192.168.13.3 remote-as 23
R1(config-router)#neighbor 192.168.14.4 remote-as 45
R1(config-router)#neighbor 192.168.12.2 remote-as 23
R1(config-router)#exit
R1(config)#
```

*Redistributions ospf dans bgp

```
R1(config-router)#exit
R1(config)#router bgp 1
R1(config-router)#redistribute ospf 1 match external internal
R1(config-router)#exit
R1(config)#
```

Redistribution bgp dans ospf

```
R1(config)#router ospf 1
R1(config-router)#redistribute bgp 23 subnets
BGP is already running; AS is 1
R1(config-router)#redistribute bgp 45 subnets
BGP is already running; AS is 1
R1(config-router)#redistribute bgp 999 subnets
BGP is already running; AS is 1
R1(config-router)#redistribute bgp 999 subnets
BGP is already running; AS is 1
R1(config-router)#
```

Table de routage

R1

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.12.0/24 is directly connected, FastEthernet1/0
C    192.168.13.0/24 is directly connected, FastEthernet0/1
     203.0.113.0/30 is subnetted, 1 subnets
C      203.0.113.0 is directly connected, FastEthernet0/0
C    192.168.14.0/24 is directly connected, FastEthernet1/1
R1#
```

Faisons la commande `show ip bgp summary`


```
R1#sh ip bgp summary
BGP router identifier 203.0.113.1, local AS number 1
BGP table version is 1, main routing table version 1
1 network entries using 117 bytes of memory
1 path entries using 52 bytes of memory
2/0 BGP path/bestpath attribute entries using 248 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 417 total bytes of memory
BGP activity 1/0 prefixes, 1/0 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
192.168.12.2   4    23      0      0        0    0    0 never    Active
192.168.13.3   4    23      0      0        0    0    0 never    Active
192.168.14.4   4    45      0      0        0    0    0 never    Active
203.0.113.2    4   999      0      0        0    0    0 never    Active
R1#
```

As999

Objectif : Sur le **routeur ISP (AS 999)**, nous **devons configurer le protocole BGP** pour établir une session avec l'AS1 (R1). **BGP est le standard** pour l'interconnexion entre AS (ISP ↔ Client)

Configuration

Adressage ip

```
ISP#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ISP(config)#ip add 203.0.113.2 255.255.255.0
      ^
% Invalid input detected at '^' marker.

ISP(config)#int fa0/0
ISP(config-if)#ip add 203.0.113.2 255.255.255.0
ISP(config-if)#no sh
ISP(config-if)#exit
*Mar  1 00:18:01.163: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar  1 00:18:02.163: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
ISP(config-if)#exit
ISP(config)#
```

Configuration protocole bgp

```
ISP(config)#router bgp 999
ISP(config-router)#neighbor 203.0.113.1 remote-as 1
ISP(config-router)#network 2
*Mar  1 00:36:46.847: %BGP-5-ADJCHANGE: neighbor 203.0.113.1 Up
ISP(config-router)#exit
ISP(config)#
```

AS23

*objectif

Dans cette AS on utilisera le protocole eigrp et le protocole bgp pour assurer la connectivité avec l'AS 1 et l'AS4567 et faire la redistribution des routes, afficher le table de routage de chaque routeurs au sein de l'AS. on s'attend á voir tous les routes de l'AS 1 annoncé via bgp.

Configuration routeurs

R2

Adressage ip

Vers R1

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#int fa0/0
R2(config-if)#ip add 192.168.12.2 255.255.255.0
R2(config-if)#no sh
R2(config-if)#exit
R2(config)#
*Mar  1 00:25:36.659: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar  1 00:25:37.659: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R2(config)#
```

Projet technologie internet

p

Vers R3

```
R2(config)#int fa0/1
R2(config-if)#ip add 192.168.23.2 255.255.255.0
R2(config-if)#no sh
R2(config-if)#exit
R2(config)#
*Mar  1 00:27:30.171: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar  1 00:27:31.171: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1
, changed state to up
R2(config)#
```

Vers R6

```
R2(config)#int fal/0
R2(config-if)#ip add 192.168.24.2 255.255.255.0

% IP addresses may not be configured on L2 links.

R2(config-if)#no switchport
R2(config-if)#ip add 192.168.24.2 255.255.255.0
*Mar  1 00:29:11.235: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed
state to down
R2(config-if)#ip add 192.168.24.2 255.255.255.0
R2(config-if)#
*Mar  1 00:29:13.391: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0
, changed state to up
R2(config-if)#no sh
R2(config-if)#exit
R2(config)#
```

Configuration protocole eigrp

```
R2(config)#router eigrp 23
R2(config-router)#network 192.168.23.0 0.0.0.255
R2(config-router)#exit
R2(config)#
```

Configuration protocole bgp

```
R2(config)#router bgp 23
R2(config-router)#neighbor 192.168.12.1 remote-as 1
R2(config-router)#neighbor 192.168.24.4 remote-as 45
*Mar  1 00:36:08.731: %BGP-5-ADJCHANGE: neighbor 192.168.12.1 Up
R2(config-router)#neighbor 192.168.24.4 remote-as 4567
R2(config-router)#neighbor 192.168.24.4 remote-as 45
R2(config-router)#exit
R2(config)#
```

Redistribution de eigrp dans bgp

```
R2(config)#router bgp 23
R2(config-router)#redistribute eigrp 23
R2(config-router)#exit
R2(config)#
```

Redistribution de bgp dans eigrp

```
R2(config)#router eigrp 23
R2(config-router)#redistribute bgp 23 metric 1000 1 1 14000
^
% Invalid input detected at '^' marker.

R2(config-router)#redistribute bgp 23 metric 1000 1 1 1400
^
% Invalid input detected at '^' marker.

R2(config-router)#redistribute bgp 23 metric 1000 1 1 140
% Incomplete command.

R2(config-router)#redistribute bgp 23
R2(config-router)#redistribute bgp 1
BGP is already running; AS is 23
R2(config-router)#redistribute bgp 4567
BGP is already running; AS is 23
R2(config-router)#redistribute bgp 45
BGP is already running; AS is 23
R2(config-router)#
```

R3

Adressage ip

Projet technologie internet

p

Vers R1

```
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#int fa0/0
R3(config-if)#ip add 192.168.13.3 255.255.255.0
R3(config-if)#no sh
R3(config-if)#exit
R3(config)#
*Mar 1 00:34:31.223: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar 1 00:34:32.223: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R3(config)#
```

Vers R2

```
R3(config)#int fa0/1
R3(config-if)#ip add 192.168.23.3 255.255.255.0
R3(config-if)#no sh
R3(config-if)#exit
R3(config)#
*Mar 1 00:36:11.879: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:36:12.879: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up
R3(config)#
```

Vers R5

```
R3(config)#int fal/0
R3(config-if)#ip add 192.168.35.3 255.255.255.0

% IP addresses may not be configured on L2 links.

R3(config-if)#no sw
R3(config-if)#ip add 192.168.35.3 255.255.255.0
*Mar 1 00:37:52.491: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed s
tate to down
R3(config-if)#ip add 192.168.35.3 255.255.255.0
R3(config-if)#
*Mar 1 00:37:54.647: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0,
changed state to up
R3(config-if)#no sh
R3(config-if)#exit
R3(config)#
```

Configuration eigrp

```
R3(config)#router eigrp 23
R3(config-router)#network 192.168.23.0 0.0.0.255
R3(config-router)#
*Mar  1 00:40:37.915: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 23: Neighbor 192.168.23.2 (FastEthernet0/1) is up: new adjacency
R3(config-router)#exit
R3(config)#
```

Configuration bgp

```
R3(config)#router bgp 23
R3(config-router)#neighbor 192.168.13.1 remote-as 1
R3(config-router)#neighbor 192.168.13.1 remote-as 1
*Mar  1 00:43:36.131: %BGP-5-ADJCHANGE: neighbor 192.168.13.1 Up
R3(config-router)#neighbor 192.168.35.5 remote-as 4567
R3(config-router)#neighbor 192.168.35.5 remote-as 45
R3(config-router)#exit
R3(config)#
```

Redistribution eigrp dans bgp

```
R3(config)#router bgp 23
R3(config-router)#redistribute eigrp 23
R3(config-router)#exit
R3(config)#
```

Redistribution de bgp dans eigrp

```
R3(config)#router eigrp 23
R3(config-router)#redistribute bgp 1 metric 1 1 1 1200
^
% Invalid input detected at '^' marker.

R3(config-router)#redistribute bgp 1
BGP is already running; AS is 23
R3(config-router)#redistribute bgp 4567
BGP is already running; AS is 23
R3(config-router)#redistribute bgp 45
BGP is already running; AS is 23
R3(config-router)#redistribute bgp 23
R3(config-router)#
R3#
*Mar  1 00:50:19.971: %SYS-5-CONFIG_I: Configured from console by console
R3#
```

Table de routage

R2

```

R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.12.0/24 is directly connected, FastEthernet0/0
C    203.0.113.0/30 is subnetted, 1 subnets
B      203.0.113.0 [20/0] via 192.168.12.1, 00:27:28
C    192.168.24.0/24 is directly connected, FastEthernet1/0
C    192.168.23.0/24 is directly connected, FastEthernet0/1
R2#
```

R3

```

R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.13.0/24 is directly connected, FastEthernet0/0
C    203.0.113.0/30 is subnetted, 1 subnets
B      203.0.113.0 [20/0] via 192.168.13.1, 00:09:19
C    192.168.23.0/24 is directly connected, FastEthernet0/1
C    192.168.35.0/24 is directly connected, FastEthernet1/0
R3#
```

Remarque : on parvient à voir tous les routes de l'AS1 et celui de l'AS 4567 grace á la redistribution du bgp dans eigrp qui a redistribué les routes apprises

AS4567

➤ AS45

Objectif

Dans cette As on utilisera le protocole rip _et le protocole bgp pour assurer la connectivité avec l'As 23 et les autres AS et faire la redistribution des routes , afficher le table de routage de chaque routeurs au sein de l'AS. On s'attend á voir tous les réseaux de l'AS23 qui sera apprises via bgp.

Configuration routeurs

R4

*Adressage ip

*Vers R1

```
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#int fa0/0
R4(config-if)#ip add 192.168.14.4 255.255.255.0
R4(config-if)#no sh
R4(config-if)#exit
R4(config)#
*Mar  1 00:47:57.243: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar  1 00:47:58.243: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R4(config)#
```


*Vers R2

```
R4(config)#int fal/1
R4(config-if)#ip add 192.168.24.4 255.255.255.0

% IP addresses may not be configured on L2 links.

R4(config-if)#ip add 192.168.24.4 255.255.255.0

% IP addresses may not be configured on L2 links.

R4(config-if)#no sw
R4(config-if)#ip add 192.168.24.4 255.255.255.0
R4(config-if)#no
*Mar 1 00:50:48.115: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/1, changed state to up
R4(config-if)#no sh
R4(config-if)#exit
R4(config)#
```

*Vers R5

```
R4(config)#int fa0/1
R4(config-if)#ip add 192.168.45.4 255.255.255.0
R4(config-if)#no sh
R4(config-if)#
*Mar 1 00:52:32.939: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:52:33.939: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R4(config-if)#
```

Vers R6

```
R4(config)#int fal/0
R4(config-if)#ip add 192.168.46.4 255.255.255.0

% IP addresses may not be configured on L2 links.

R4(config-if)#np sw
^
% Invalid input detected at '^' marker.

R4(config-if)#no sw
R4(config-if)#ip add 192.168.46.4 255.255.255.0
*Mar 1 00:54:27.371: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to down
R4(config-if)#ip add 192.168.46.4 255.255.255.0
*Mar 1 00:54:29.527: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R4(config-if)#ip add 192.168.46.4 255.255.255.0
R4(config-if)#no sh
R4(config-if)#exit
R4(config)#
```

Configuration du protocole RIP

```
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#network 192.168.46.0
R4(config-router)#network 192.168.45.0
R4(config-router)#
```

Configuration bgp avec R4

```
R4(config)#router bgp 4567
R4(config-router)#neighbor 192.168.14.1 remote-as 1
R4(config-router)#neighbor 192.168.24.2 remote-as 23
R4(config-router)#
*Mar 1 01:03:25.639: %BGP-3-NOTIFICATION: received from neighbor 192.168.14.1 2/2
(peer in wrong AS) 2 bytes 11D7
R4(config-router)#EXIT
*Mar 1 01:03:33.211: %BGP-3-NOTIFICATION: received from neighbor 192.168.24.2 2/2
(peer in wrong AS) 2 bytes 11D7
R4(config-router)#EXIT
R4(config)#
```

Redistribution rip dans bgp

```
R4(config)#router bgp 4567
R4(config-router)#router bgp 45
BGP is already running; AS is 4567
R4(config)#router bgp 4567
R4(config-router)#
*Mar 1 01:04:52.275: %BGP-3-NOTIFICATION: received from neighbor 192.168.24.2 2/2
(peer in wrong AS) 2 bytes 11D7
R4(config-router)#
*Mar 1 01:04:55.351: %BGP-3-NOTIFICATION: received from neighbor 192.168.14.1 2/2
(peer in wrong AS) 2 bytes 11D7
R4(config-router)#redistribute rip
R4(config-router)#
*Mar 1 01:05:21.567: %BGP-3-NOTIFICATION: received from neighbor 192.168.24.2 2/2
(peer in wrong AS) 2 bytes 11D7
R4(config-router)#
*Mar 1 01:05:26.963: %BGP-3-NOTIFICATION: received from neighbor 192.168.14.1 2/2
(peer in wrong AS) 2 bytes 11D7
R4(config-router)#
```

Redistribution de bgp dans rip

Projet technologie internet

p

```
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#
*Mar  1 01:06:54.831: %BGP-3-NOTIFICATION: received from neighbor 192.168.14.1 2/2
(peer in wrong AS) 2 bytes l1D7
R4(config-router)#redistribute bgp 45
*Mar  1 01:07:13.623: %BGP-3-NOTIFICATION: received from neighbor 192.168.24.2 2/2
(peer in wrong AS) 2 bytes l1D7
R4(config-router)#redistribute bgp 45
BGP is already running; AS is 4567
R4(config-router)#redistribute bgp 4567
R4(config-router)#redistribute bgp 4567
*Mar  1 01:07:23.275: %BGP-3-NOTIFICATION: received from neighbor 192.168.14.1 2/2
(peer in wrong AS) 2 bytes l1D7
R4(config-router)#redistribute bgp 23
BGP is already running; AS is 4567
R4(config-router)#
*Mar  1 01:07:40.987: %BGP-3-NOTIFICATION: received from neighbor 192.168.24.2 2/2
(peer in wrong AS) 2 bytes l1D7
R4(config-router)#exit
R4(config)#
```

R5

*adressage ip

*Vers R3

```
R5(config)#int fa0/0
R5(config-if)#ip add 192.168.35.5 255.255.255.0
R5(config-if)#no sh
R5(config-if)#
*Mar  1 00:57:28.279: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar  1 00:57:29.279: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R5(config-if)#exit
R5(config)#
```

*Vers R4

```
R5(config)#int fa0/1
R5(config-if)#ip add 192.168.45.5 255.255.255.0
R5(config-if)#no sh
R5(config-if)#exit
R5(config)#
*Mar  1 00:58:57.579: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state t
o up
*Mar  1 00:58:58.579: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/1, changed state to up
R5(config)#
```

Vers R7

```
R5(config)#int fal/0
R5(config-if)#ip add 192.168.57.5 255.255.255.0

% IP addresses may not be configured on L2 links.

R5(config-if)#no sw
R5(config-if)#ip add 192.168.57.5 255.255.255.0
R5(config-if)#ip add 192.168.57.5 255.255.255.0
*Mar  1 01:00:41.415: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/
0, changed state to up
R5(config-if)#no sh
R5(config-if)#exit
R5(config)#
```

Vers R8

```
R5(config)#int fal/1
R5(config-if)#ip add 192.168.58.5 255.255.255.0

% IP addresses may not be configured on L2 links.

R5(config-if)#NO SW
R5(config-if)#ip add 192.168.58.5 255.255.255.0
*Mar  1 01:03:38.599: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed
state to down
R5(config-if)#ip add 192.168.58.5 255.255.255.0
R5(config-if)#
R5(config-if)#
*Mar  1 01:03:40.755: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/
1, changed state to up
R5(config-if)#no sh
R5(config-if)#exit
R5(config)#
```

Configuration rip

```
R5(config)#router rip
R5(config-router)#
R5(config-router)#version 2
R5(config-router)#net 192.168.57.0
R5(config-router)#net 192.168.45.0
R5(config-router)#
```

Configuration bgp

```
R5(config-router)#router bgp 4567
R5(config-router)#neighbor 192.168.35.3 remote-as 23
R5(config-router)#neighbor 192.168.58.3 remote-as 23
*Mar  1 01:09:43.147: %BGP-3-NOTIFICATION: received from neighbor 192.168.35.3 2/2 (
peer in wrong AS) 2 bytes 11D7
R5(config-router)#neighbor 192.168.58.8
% Incomplete command.

R5(config-router)#
*Mar  1 01:10:10.351: %BGP-3-NOTIFICATION: received from neighbor 192.168.35.3 2/2 (
peer in wrong AS) 2 bytes 11D7
R5(config-router)#
```

Redistribution rip dans bgp

```
R5(config-router)#router bgp 4567
R5(config-router)#re
*Mar  1 01:11:35.747: %BGP-3-NOTIFICATION: received from neighbor 192.168.35.3 2/2 (
peer in wrong AS) 2 bytes 11D7
R5(config-router)#redistribute rip
R5(config-router)#
```

Redistribution de bgp dans rip

```
R5(config)#router rip
R5(config-router)#version 2
R5(config-router)#redistribute
*Mar  1 01:13:05.935: %BGP-3-NOTIFICATION: received from neighbor 192.168.35.3 2/2 (
peer in wrong AS) 2 bytes 11D7
R5(config-router)#redistribute bgp 45
BGP is already running; AS is 4567
R5(config-router)#redistribute bgp 4567
R5(config-router)#redistribute bgp 45
*Mar  1 01:13:31.683: %BGP-3-NOTIFICATION: received from neighbor 192.168.35.3 2/2 (
peer in wrong AS) 2 bytes 11D7
R5(config-router)#redistribute bgp 23
BGP is already running; AS is 4567
R5(config-router)#exxit
^
% Invalid input detected at '^' marker.

R5(config-router)#exit
R5(config)#
```

➤ Sub-AS67

Objectif

Dans cette Sub-AS comme dit précédemment on va utiliser le protocole ospf au sein du SUB-AS67 et le protocole bgp pour assurer la connectivité avec le SUB-AS 45 , faire la redistribution des routes et afficher la table de routage de chaque routeurs de l'AS.

A- Configurations routeurs

R6

Adressage ip

vert R4

```
R6(config)#int fa0/0
R6(config-if)#ip add 192.168.46.6 255.255.255.0
R6(config-if)#no sh
R6(config-if)#exit
R6(config)#
*Mar  1 01:05:23.667: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar  1 01:05:24.667: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R6(config)#
```

vert R7

```
ig)#int fa0/1
ig-if)#ip add 192.168.67.6 255.255.255.0
ig-if)#no sh
ig-if)#exit
ig)#
 01:07:19.835: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state t
 01:07:20.835: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
changed state to up
ig)#
```

Configuration ospf

```
et0/1, changed state to up
R6(config)#router ospf 67
R6(config-router)#router ospf 4567
R6(config-router)#network 192.168.67.0 0.0.0.255 area 0
R6(config-router)#network 192.168.46.0 0.0.0.255 area 0
R6(config-router)#
```

Configuration de bgp

```
R6(config)#router bgp 4567
R6(config-router)#router bgp 67
BGP is already running; AS is 4567
R6(config)#router bgp 4567
R6(config-router)#neighbor 192.168.46.4 remote-as 45
R6(config-router)#
```

Redistribution de ospf dans bgp

```
R6(config)#router bgp 4567
R6(config-router)#redistribute ospf 1 match internal external
                                     ^
% Invalid input detected at '^' marker.

R6(config-router)#redistribute ospf 1 match internal external
R6(config-router)#redistribute ospf 1 match external internal
R6(config-router)#exit
R6(config)#
```

Redistribution de bgp dans ospf

```
R6(config)#router ospf 67
R6(config-router)#router ospf 4567
R6(config-router)#redistribute bgp 67 subnets
BGP is already running; AS is 4567
R6(config-router)#redistribute bgp 4567 subnets
R6(config-router)#redistribute bgp 23 subnets
BGP is already running; AS is 4567
R6(config-router)#exit
R6(config)#
```


R7

Adressage ip

vert R5

```
R7(config)#int fa0/1
R7(config-if)#ip add 192.168.57.7 255.255.255.0
R7(config-if)#no sh
R7(config-if)#exit
R7(config)#
*Mar  1 01:13:27.527: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state t
o up
*Mar  1 01:13:28.527: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/1, changed state to up
R7(config)#
```

vert R6

```
R7(config)#int fa0/0
R7(config-if)#ip add 192.168.67.7 255.255.255.0
R7(config-if)#no sh
R7(config-if)#exit
R7(config)#
*Mar  1 01:14:32.683: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar  1 01:14:33.683: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R7(config)#
```

Configuration protocole Ospf

```
R7(config)#router ospf 67
R7(config-router)#router ospf 4567
R7(config-router)#network 192.168.67.0 0.0.0.255 area 0
R7(config-router)#network 192.168.57.0 0.0.0.255 area 0
*Mar  1 01:17:06.551: %OSPF-5-ADJCHG: Process 4567, Nbr 192.168.46.6 on FastEthe
rnet0/0 from LOADING to FULL, Loading Done
R7(config-router)#network 192.168.57.0 0.0.0.255 area 0
R7(config-router)#
```

Configuration protocole bgp

```
R7(config)#router bgp 67
R7(config-router)#router bgp 4567
BGP is already running; AS is 67
R7(config)#router bgp 67
R7(config-router)#neighbor 192.168.57.5 remote-as 45
R7(config-router)#
```

Redistribution de ospf dans bgp


```
R7(config)#router bgp 67
R7(config-router)#router bgp 4567
BGP is already running; AS is 67
R7(config)#router bgp 67
R7(config-router)#redistribute ospf 67 match internal external
R7(config-router)#exit
```

Redistribution de bgp dans ospf

```
R7(config)#router ospf 67
R7(config-router)#
R7(config-router)#redistribute bgp 67 subnets
R7(config-router)#redistribute bgp 23 subnets
BGP is already running; AS is 67
R7(config-router)#redistribute bgp 4567 subnets
BGP is already running; AS is 67
R7(config-router)#
```

Table de routage

R6

```
R6#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.46.0/24 is directly connected, FastEthernet0/0
O    192.168.57.0/24 [110/20] via 192.168.67.7, 00:07:07, FastEthernet0/1
C    192.168.67.0/24 is directly connected, FastEthernet0/1
R6#
```

R7

```
R7#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

O    192.168.46.0/24 [110/20] via 192.168.67.6, 00:09:05, FastEthernet0/0
C    192.168.57.0/24 is directly connected, FastEthernet0/1
C    192.168.67.0/24 is directly connected, FastEthernet0/0
R7#
```

**On parvient à voir tous les routes de l'As 23 et l'As4567
appries via bgp !!!!!**

As 65505

Objectif : AS65505 sert de **point de sortie/entrée sécurisé** pour le trafic externe, avec des politiques BGP strictes pour éviter les annonces de routes non autorisées. Dans cet AS, on va configurer le protocole de routage ospf pour la coherence avec d'autre AS pour les zone intra AS ainsi que le protocole de routage BGP pour les zones inter AS

Configuration

R9

Adressage ip

Avec R8

```
R9(config)#int fa0/0
R9(config-if)#ip add 192.168.89.9 255.255.255.0
R9(config-if)#no sh
R9(config-if)#
*Mar  1 01:22:51.631: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar  1 01:22:52.631: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R9(config-if)#exit
R9(config)#
```

Configuration ospf

```
R9(config)#router ospf 65505
R9(config-router)#exit
R9(config)#
```

Configuration bgp

```
R9(config)#router bgp 65505
R9(config-router)#neighbor 192.168.89.8
```

Redistribution Ospf dans bgp

```
R9(config)#router bgp 65505
R9(config-router)#redistribute ospf 65505 match external internal
R9(config-router)#exit
```

Redistribution bgp dans ospf

```
R9(config)#router ospf 65505
R9(config-router)#redistribute bgp 4567 subnets
BGP is already running; AS is 65505
R9(config-router)#exit
R9(config)#
```

R8

Adressage ip

Avec R9

```
R8(config)#int fa0/0
R8(config-if)#ip add 192.168.89.8 255.255.255.0
R8(config-if)#no sh
R8(config-if)#exit
R8(config)#
*Mar  1 01:33:18.219: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar  1 01:33:19.219: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R8(config)#
```

Avec R5

```
R8(config)#int fa0/1
R8(config-if)#ip add 192.168.58.8 255.255.255.0
R8(config-if)#no sh
R8(config-if)#exit
R8(config)#
*Mar  1 01:34:38.743: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state t
o up
*Mar  1 01:34:39.743: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/1, changed state to up
R8(config)#
```

II. Mise en place des services reseaux

• DHCP :

- Sur **R1 (AS1)** pour attribuer des IP aux clients du LAN (203.0.113.0/20)

Le **protocole DHCP (Dynamic Host Configuration Protocol)** est un service essentiel dans ce projet de réseau multi-AS, car il simplifie la gestion des adresses IP et des paramètres réseau pour les appareils connectés. Voici son rôle détaillé dans l'architecture. Elle permet une attribution automatique des adresses ip vers les machines des utilisateurs. Dans le cas de notre projet, l'AS 1 est relie a

un fournisseur d'Access a internet. Donc on va configurer le protocole DHCP sur le routeur 1 de l'AS1

Conf t ; Configuration du terminal

Service dhcp ; Pour répondre aux requetes DHCP et activer le processus du serveur dhcp

ip dhcp pool LANPOOL : Pour configurer le nom du pool de LANPOOL

dns-server [@ip_du_serveur_dns] ; Pour repondre aux requetes DHCP et activer le processus du serveur dhcp

ip dhcp excluded-address [@ip_initial -@ip_final] : Pour exclure des address afin d'éviter les conflits d'address

network [@ip_du_reseaux] [mask_du_reseaux] : Attribut les address ip aux clients

lease 2 : loue chaque add ip pendant 2 jours

default-router [@ip_du_routeur] : Utilise l'address ip de fastethernet comme passerelle par default pour les clients DHCP de ce pool

show ip dhcp binding : Verifier la fonctionnalite du serveur dhcp

sh ip dhcp server statistics : Donne les résultats généraux des statistiques

sh ip arp : Vérifie la table de routage du serveur DHCP NAT

Projet technologie internet

p

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
*Mar 1 04:08:06.042: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer in wrong AS) 2 b
ytes 11D7
R1(config)# FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 COA8 2E04 1002 0601 0400 01
00 0102 0280 0002 0202 00
R1(config)#service dhcp
R1(config)#ip dhcp pool
*Mar 1 04:08:35.658: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer in wrong AS) 2 b
ytes 11D7
R1(config)#ip dhcp pool FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 COA8 2E04 1002
0601 0400 0100 0102 0280 0002 0202 00
R1(config)#ip dhcp pool LANPOOL
R1(dhcp-config)#network 203.0.113.0
*Mar 1 04:09:08.490: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer in wrong AS) 2 b
ytes 11D7
R1(dhcp-config)#network 203.0.113.0 FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 COA
8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00
R1(dhcp-config)#network 203.0.113.0 255.255.255.0
R1(dhcp-config)#lease 2
R1(dhcp-config)#default-router
*Mar 1 04:09:39.586: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer in wrong AS) 2 b
ytes 11D7
R1(dhcp-config)#default-router FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 COA8 2E0
4 1002 0601 0400 0100 0102 0280 0002 0202 00
R1(dhcp-config)#default-router 203.0.113.
*Mar 1 04:10:09.778: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer in wrong AS) 2 b
ytes 11D7
R1(dhcp-config)#default-router 203.0.113. FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00
B4 COA8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00
R1(dhcp-config)#default-router 203.0.113.2
R1(dhcp-config)#dns=-serveur 203.0.113
*Mar 1 04:10:43.030: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer in wrong AS) 2 b
ytes 11D7
R1(dhcp-config)#dns=-serveur 203.0.113. FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4
COA8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00
```

Exécutons la commande **sh ip dhcp binding**

```
R1#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/          Lease expiration    Type
                   Hardware address/
                   User name
R1#
*Mar 1 04:13:08.070: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer in wrong AS) 2 b
ytes 11D7
R1# FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 COA8 2E04 1002 0601 0400 0100 0102
0280 0002 0202 00
```

Exécutons la commande **sh ip dhcp server statistics**

074

```
sh ip arp
```

R1#

- **NAT :**

- Sur **R1** pour permettre au LAN d'accéder à Internet via l'ISP (AS999).

Le **NAT (Network Address Translation)** joue un rôle crucial dans ce projet en permettant aux appareils du réseau privé de l'**AS1** de communiquer avec Internet via l'**ISP (AS999)**

Le NAT traduit les adresses privées en une **adresse publique** (fournie par l'ISP) pour les communications externes.

sh ip nat translation Permet de vérifier les translations

Donc configurons le NAT sur le Routeur R1 de l'AS 1

```
R1(config-if)#int fa0/0
R1(config-if)#i
*Mar  1 04:44:37.678: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer in wrong AS) 2 bytes 11D7
R1(config-if)#ip  FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 C0A8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00
R1(config-if)#ip nat inside
R1(config-if)#
*Mar  1 04:44:45.442: %LINEPROTO-5-UPDOWN: Line protocol on Interface NVI0, changed state to up
R1(config-if)#int fa0/1
R1(config-if)#ip nat inside
R1(config-if)#int fa
*Mar  1 04:45:10.730: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer in wrong AS) 2 bytes 11D7
R1(config-if)#int fa FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 C0A8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00
R1(config-if)#int fa/0
R1(config-if)#ip nat inside
R1(config-if)#int fa/1
R1(config-if)#ip nat inside
R1(config-if)#int fa/2
R1(config-if)#ip nat outside
R1(config-if)#
^
% Invalid input detected at '^' marker.

R1(config-if)#
*Mar  1 04:45:37.998: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer in wrong AS) 2 bytes 11D7
R1(config-if)# FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 C0A8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00
R1(config-if)#
```

Ensuite on met

#ip nat inside source static 192.168.13.1 192.168.13.3

#ip nat outside source static 192.168.13.1 203.0.103.2

- **DNS :**

- Serveur DNS dans **AS4567** pour résoudre les noms entre AS.

DNS (Domain Name System) est un système essentiel de l'internet qui agit comme un "annuaire" pour traduire les **noms de domaines** (comme google.com) en **adresses IP** (comme 172.217.14.206), compréhensibles par les machines. Ainsi nous allons configurer notre DNS sur l'AS 4567

Nous devons configurer le serveur DNS sur **un routeur central de l'AS4567**, idéalement sur **R4** (appartenant au **Sub-AS45**), car ce routeur est connecté aux autres AS et garantissent une accessibilité optimale car un serveur DNS centralisé dans l'AS4567 simplifie la gestion et réduit la latence

Sub-AS 45

R4

```
ip dns server // Active le serveur DNS sur le routeur
ip host server.as1.sn 192.168.14.1 // Hôte dans AS1
ip host server.as23.sn 192.168.24.2 // Hôte dans AS23
ip host server.as65505.sn 192.168.45.4 // Passerelle de l'AS65505
```

Projet technologie internet

p

```
R4(config)#ip dns server
R4(config)#
*Mar 1 04:35:35.562: %BGP-3-NOTIFICATION: received from neighbor 192.168.24.2 2/2 (peer in wrong AS) 2 bytes 11D7
R4(config)#ip host
*Mar 1 04:35:43.930: %BGP-3-NOTIFICATION: received from neighbor 192.168.14.1 2/2 (peer in wrong AS) 2 bytes 11D7
R4(config)#ip host server
*Mar 1 04:36:06.146: %BGP-3-NOTIFICATION: received from neighbor 192.168.24.2 2/2 (peer in wrong AS) 2 bytes 11D7
R4(config)#ip host server.as1.
*Mar 1 04:36:15.966: %BGP-3-NOTIFICATION: received from neighbor 192.168.14.1 2/2 (peer in wrong AS) 2 bytes 11D7
R4(config)#ip host server.as1.sn
*Mar 1 04:36:32.778: %BGP-3-NOTIFICATION: received from neighbor 192.168.24.2 2/2 (peer in wrong AS) 2 bytes 11D7
R4(config)#ip host server.as1.sn 192.1
*Mar 1 04:36:47.438: %BGP-3-NOTIFICATION: received from neighbor 192.168.14.1 2/2 (peer in wrong AS) 2 bytes 11D7
R4(config)#ip host server.as1.sn 192.168.14.1
R4(config)#
*Mar 1 04:36:58.914: %BGP-3-NOTIFICATION: received from neighbor 192.168.24.2 2/2 (peer in wrong AS) 2 bytes 11D7
R4(config)#ip host server.as23.sn 192.168.14.1
*Mar 1 04:37:13.206: %BGP-3-NOTIFICATION: received from neighbor 192.168.14.1 2/2 (peer in wrong AS) 2 bytes 11D7
R4(config)#ip host server.as23.sn 192.168.24.1
*Mar 1 04:37:27.054: %BGP-3-NOTIFICATION: received from neighbor 192.168.24.2 2/2 (peer in wrong AS) 2 bytes 11D7
R4(config)#ip host server.as23.sn 192.168.24.2
R4(config)#ip host server.as23.sn 192.168.24.2
*Mar 1 04:37:43.602: %BGP-3-NOTIFICATION: received from neighbor 192.168.14.1 2/2 (peer in wrong AS) 2 bytes 11D7
R4(config)#ip host server.as23.sn 192.168.45.5
R4(config)#
*Mar 1 04:37:55.262: %BGP-3-NOTIFICATION: received from neighbor 192.168.24.2 2/2 (peer in wrong AS) 2 bytes 11D7
R4(config)#exit
R4#
*Mar 1 04:38:04.314: %SYS-5-CONFIG_I: Configured from console by console
R4#
*Mar 1 04:38:12.958: %BGP-3-NOTIFICATION: received from neighbor 192.168.14.1 2/2 (peer in wrong AS) 2 bytes 11D7
R4#
```

```
R4(config)#ip host server.as65505.sn 192.168.45.5
*Mar 1 04:59:29.418: %BGP-3-NOTIFICATION: received from neighbor 192.168.24.2 2/2 (peer in wrong AS) 2 bytes 11D7
R4(config)#ip host server.as65505.sn 192.168.45.5
R4(config)#
*Mar 1 04:59:38.090: %BGP-3-NOTIFICATION: received from neighbor 192.168.14.1 2/2 (peer in wrong AS) 2 bytes 11D7
R4(config)#ip host server.as999.sn 192.1
*Mar 1 04:59:59.926: %BGP-3-NOTIFICATION: received from neighbor 192.168.24.2 2/2 (peer in wrong AS) 2 bytes 11D7
R4(config)#ip host server.as999.sn 203.0.113.2
R4(config)#exit
R4#
*Mar 1 05:00:10.658: %BGP-3-NOTIFICATION: received from neighbor 192.168.14.1 2/2 (peer in wrong AS) 2 bytes 11D7
R4#
*Mar 1 05:00:11.994: %SYS-5-CONFIG_I: Configured from console by console
R4#
```

```

R1(dhcp-config)#dns-server 192.168.14.4
R1(dhcp-config)#exit
*Mar 1 05:11:53.690: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer in wrong AS) 2 bytes 11D7
R1(dhcp-config)#exit
R1(config)# FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 C0A8 2E04 1
002 0601 0400 0100 0102 0280 0002 0202 00
```

Projet technologie internet

p

```
7 00E4 C0A8 2E04 1002 0601 0400 0100 0102 0200 0002 0202 00
R2(config)#ip dhcp pool LANPOOL
R2(dhcp-config)#dns-server 192.166.14.4
R2(dhcp-config)#
```

```
8 1002 0002 0100 0102 0200 0002 0202 00
R3(config)#ip dhcp pool LANPOOL
R3(dhcp-config)#dns-server 192.168.14.4
R3(dhcp-config)#
```

```
9 2 (peer in wrong AS) 2 bytes 11D7
R5(config)#ip dhcp pool LANPOOL
R5(dhcp-config)#dns server
*Mar 1 04:47:09.686: %BGP-3-NOTIFICATION: received from neighbor 192.168.3
/2 (peer in wrong AS) 2 bytes 11D7
R5(dhcp-config)#dns server 192.168.14.4
Translating "server"
^
% Invalid input detected at '^' marker.

R5(dhcp-config)#dns server 192.168.14.4
Translating "server"
^
% Invalid input detected at '^' marker.

R5(dhcp-config)#
*Mar 1 04:47:37.526: %BGP-3-NOTIFICATION: received from neighbor 192.168.3
/2 (peer in wrong AS) 2 bytes 11D7
R5(dhcp-config)#dns-server 192.168.14.4
R5(dhcp-config)#
```

```
10 (peer in wrong AS) 2 bytes 11D7
R6(config)#ip dhcp pool LANPOOL
R6(dhcp-config)#dns-server 192.168.14.4
R6(dhcp-config)#
```

```
11 (peer in wrong AS) 2 bytes 11D7
R7(config)#ip dhcp pool LANPOOL
R7(dhcp-config)#dns-server 192.168.14.4
R7(dhcp-config)#
```

```
12
R9(config)#ip dhcp pool LANPOOL
R9(dhcp-config)#dns-server
% Incomplete command.

R9(dhcp-config)#dns-server 192.168.14.4
R9(dhcp-config)#
```

```
13
R8(config)#ip dhcp pool LANPOOL
R8(dhcp-config)#dns-server 192,.168.14.4
^
% Invalid input detected at '^' marker.

R8(dhcp-config)#dns-server 192.168.14.4
R8(dhcp-config)#
```

III. Securisation des réseaux

Les **ACLs (Access Control Lists)** jouent un rôle clé dans la sécurisation et la gestion du trafic au sein de ce réseau multi-AS

ACL :

- a. Bloquer le trafic indésirable (ex: ICMP entre AS1 et AS23).
- b. Filtrer les accès depuis l'ISP (AS999).

R1

ip prefix-list FILTER AS23 seq 5 permit
192.168.12.0/24

ip prefix-list FILTER AS23 seq 5 permit
192.168.13.0/24

route-map FILTER AS23 permit 10

match ip address prefix-list FILTER AS23

router bgp 1

neighbor 192.168.13.3 route-map FILTER AS23 in

neighbor 192.168.12.2 route-map FILTER AS23 in

Ajouter une ACL sur R1 pour filtrer le trafic
entrant/sortant vers l'ISP :

access-list 999 permit tcp any any established

access-list 999 permit icmp any any echo-reply

access-list 999 deny ip any any

int fa0/0

ip access-group 999 in

Blocage du trafic ICMP entre AS1 et AS23 :

access-list 100 deny icmp any any

access-list 100 permit ip any any

int fa0/1

ip access-group 23 out

int fa1/1

ip access-group 23 out

```
l(config)# FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 C0A8 2E04 1
02 0601 0400 0100 0102 0280 0002 0202 00
l(config)#ip prefix-list FILTRE_AS23 seq 5 permit 192.168.13.0/24
Insertion failed - seq # exists with different policy: 5
l(config)#r
Mar  1 07:54:09.934: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
n wrong AS) 2 bytes 11D7
l(config)#ro FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 C0A8 2E04
1002 0601 0400 0100 0102 0280 0002 0202 00
l(config)#route-map FILTRE_AS23 permit 1
Mar  1 07:54:37.878: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
n wrong AS) 2 bytes 11D7
l(config)#route-map FILTRE_AS23 permit 1 FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 0
2D 0104 11D7 00B4 C0A8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00
l(config)#route-map FILTRE_AS23 permit 1
l(config-route-map)#match ip address prefix-list FILTRE_23
Mar  1 07:55:08.502: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
n wrong AS) 2 bytes 11D7
l(config-route-map)#match ip address prefix-list FILTRE_23 FFFF FFFF FFFF FFFF FFF
FFFF FFFF FFFF 002D 0104 11D7 00B4 C0A8 2E04 1002 0601 0400 0100 0102 0280 0002 0
02 00
l(config-route-map)#match ip address prefix-list FILTRE_23 in
l(config-route-map)#exit
l(config)#router
Mar  1 07:55:34.298: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
n wrong AS) 2 bytes 11D7
l(config)#router bg FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 CO
8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00
l(config)#router bgp 1
l(config-router)#neighbor 192.168.13.3 route-map
Mar  1 07:56:01.174: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
n wrong AS) 2 bytes 11D7
l(config-router)#neighbor 192.168.13.3 route-map FFFF FFFF FFFF FFFF FFFF FFFF FFF
FFFF 002D 0104 11D7 00B4 C0A8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00
l(config-router)#neighbor 192.168.13.3 route-map FILTRE_AS23 in
l(config-router)#neighbor 192.168.12.3 route-map FILTRE_AS23 in
Specify remote-as or peer-group commands first
l(config-router)#
Mar  1 07:56:28.166: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
n wrong AS) 2 bytes 11D7
l(config-router)# FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 C0A8
2E04 1002 0601 0400 0100 0102 0280 0002 0202 00
l(config-router)#
```

**Ajouter une ACL sur R1 pour filtrer le
trafic entrant/sortant vers l'ISP :**

```
R1(config)#access-list 999 permit tcp any established
*Mar 1 07:59:31.030: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
in wrong AS) 2 bytes 11D7
R1(config)#access-list 999 permit tcp any established FFFF FFFF FFFF FFFF FFF
F FFFF FFFF 002D 0104 11D7 00B4 C0A8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 0
0
R1(config)#access-list 999 permit tcp any established
^
% Invalid input detected at '^' marker.

R1(config)#access-list 999 permit i any established
*Mar 1 07:59:59.410: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
in wrong AS) 2 bytes 11D7
R1(config)#access-list 999 permit icmp FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D
0104 11D7 00B4 C0A8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00
R1(config)#access-list 999 permit icmp any any echo-reply
^
% Invalid input detected at '^' marker.

R1(config)#access-list 99 permit icmp any any echo-reply
*Mar 1 08:00:31.670: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
in wrong AS) 2 bytes 11D7
R1(config)#access-list FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4
C0A8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00
R1(config)#access-list 101 permit icmp any any echo-reply
R1(config)#access-list 101 permit tcp any any established
*Mar 1 08:00:58.050: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
in wrong AS) 2 bytes 11D7
R1(config)#access-list 101 permit tcp any any established
R1(config)# FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 C0A8 2E04 1
002 0601 0400 0100 0102 0280 0002 0202 00
R1(config)#access-list 101 deny ip any any
R1(config)#int f
*Mar 1 08:01:25.730: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
in wrong AS) 2 bytes 11D7
R1(config)#int fa0 FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 C0A8
2E04 1002 0601 0400 0100 0102 0280 0002 0202 00
R1(config)#int fa0/0
R1(config-if)#ip access-group 101 in
R1(config-if)#
*Mar 1 08:01:56.114: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer in wrong AS) 2 bytes 11D7
R1(config-if)# FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 C0A8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00
R1(config-if)#
```

Blocage du trafic ICMP entre AS1 et AS23 :

```
R1(config)#int fa0/0
R1(config-if)#ip access-group 101 in
R1(config-if)#
*Mar 1 08:01:56.114: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer in wrong AS) 2 bytes 11D7
R1(config-if)# FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 C0A8 2E04 1002 0601 0400 0100 0102 0280 0002 0202
R1(config-if)#
*Mar 1 08:02:21.510: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer in wrong AS) 2 bytes 11D7
R1(config-if)# FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 C0A8 2E04 1002 0601 0400 0100 0102 0280 0002 0202
R1(config-if)#
*Mar 1 08:02:48.466: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
in wrong AS) 2 bytes 11D7
R1(config-if)# FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 C0A8 2E0
4 1002 0601 0400 0100 0102 0280 0002 0202 00
R1(config-if)#access-list 100
*Mar 1 08:03:17.158: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
in wrong AS) 2 bytes 11D7
R1(config-if)#access-list 100 FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11
D7 00B4 C0A8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00
R1(config-if)#access-list 100 deny icmp any any
R1(config)#access-list 100 permit ip
*Mar 1 08:03:47.229: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
in wrong AS) 2 bytes 11D7
R1(config)#access-list 100 permit ip FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D
0104 11D7 00B4 C0A8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00
R1(config)#access-list 100 permit ip any any
R1(config)#int fa0/1
R1(config-if)#ip access-group 23 out
*Mar 1 08:04:17.617: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
in wrong AS) 2 bytes 11D7
R1(config-if)#ip access-group 23 out
R1(config-if)# FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 C0A8 2E0
4 1002 0601 0400 0100 0102 0280 0002 0202 00
R1(config-if)#int fal/1
R1(config-if)#ip access-group 23 out
R1(config-if)#
*Mar 1 08:04:43.597: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
in wrong AS) 2 bytes 11D7
R1(config-if)# FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 C0A8 2E0
4 1002 0601 0400 0100 0102 0280 0002 0202 00
R1(config-if)#
```

Protection BGP :

- Des annonces de routes avec prefix-list et route-map.
- Authentification des sessions BGP.

Sur **R3**

Filtrage BGP entre AS23 et AS4567 (sur R3) :

ip prefix-list FILTRE_AS4567 seq 10 permit 192.168.35.0/24


```
route-map FILTER_AS4567 permit 10
  match ip address prefix-list FILTER_AS4567
router bgp 23
  neighbor 192.168.13.1 route-map FILTER_AS4567 in # R4 (Sub-AS45)
  neighbor 192.168.35.5 route-map FILTER_AS4567 in # R4 (Sub-AS45)
```

Authentication MD5 :

```
router bgp 23
  neighbor 192.168.13.1 password SECURE_BGP_23
```

```
% Invalid input detected at '^' marker.

R3(config)#$ist FILTRE_AS4567 seq 10 permit 192.168.35.0/24
R3(config)#
*Mar 1 07:59:33.754: %BGP-3-NOTIFICATION: sent to neighbor 192.168.35.5 2/2 (pe
er in wrong AS) 2 bytes 11D7
R3(config)# FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 COA8 3A0
5 1002 0601 0400 0100 0102 0280 0002 0202 00
R3(config)#route-map FILTRE_AS4567 permit 10
R3(config-route-map)#match
*Mar 1 08:00:03.138: %BGP-3-NOTIFICATION: sent to neighbor 192.168.35.5 2/2 (pe
er in wrong AS) 2 bytes 11D7
R3(config-route-map)#match FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D
7 00B4 COA8 3A05 1002 0601 0400 0100 0102 0280 0002 0202 00
R3(config-route-map)#match ip address prefix-list FILTRE_AS4567
*Mar 1 08:00:29.694: %BGP-3-NOTIFICATION: sent to neighbor 192.168.35.5 2/2 (pe
er in wrong AS) 2 bytes 11D7
R3(config-route-map)#match ip address prefix-list FILTRE_AS4567 FFFF FFFF FFFF FF
FF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 COA8 3A05 1002 0601 0400 0100 0102 02
80 0002 0202 00
R3(config-route-map)#match ip address prefix-list FILTRE_AS4567
^
% Invalid input detected at '^' marker.

R3(config-route-map)#match ip address prefix-list FILTRE_AS4567
R3(config-route-map)#exit
R3(config)#router bgp 23
R3(config-router)#neighbor 1
*Mar 1 08:00:59.262: %BGP-3-NOTIFICATION: sent to neighbor 192.168.35.5 2/2 (pe
er in wrong AS) 2 bytes 11D7
R3(config-router)#neighbor 192.1 FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 01
04 11D7 00B4 COA8 3A05 1002 0601 0400 0100 0102 0280 0002 0202 00
R3(config-router)#neighbor 192.168.13.1 route-map FILTRE_AS4567 in
R3(config-router)#neighbor 192.168.13.1 route-map FILTRE_AS4567 in
*Mar 1 08:01:27.562: %BGP-3-NOTIFICATION: sent to neighbor 192.168.35.5 2/2 (pe
er in wrong AS) 2 bytes 11D7
R3(config-router)#neighbor 192.168.13.1 route-map FILTRE_AS4567 in FFFF FFFF FF
FF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 COA8 3A05 1002 0601 0400 0100 01
02 0280 0002 0202 00
R3(config-router)#neighbor 192.168.35.1 route-map FILTRE_AS4567 in
% Specify remote-as or peer-group commands first
R3(config-router)#
```

```
R3(config)#router bgp 23
R3(config-router)#
*Mar 1 08:02:54.034: %BGP-3-NOTIFICATION: sent to neighbor 192.168.35.5 2/2 (p
er in wrong AS) 2 bytes 11D7
R3(config-router)#n FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4
COA8 3A05 1002 0601 0400 0100 0102 0280 0002 0202 00
R3(config-router)#neighbor 192.168.13.1 password SECURE_BGP_23
*Mar 1 08:03:23.198: %BGP-3-NOTIFICATION: sent to neighbor 192.168.35.5 2/2 (p
er in wrong AS) 2 bytes 11D7
R3(config-router)#neighbor 192.168.13.1 password SECURE_BGP_23
R3(config-router)# FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4
OA8 3A05 1002 0601 0400 0100 0102 0280 0002 0202 00
R3(config-router)#
```

Sur R1

router bgp 1

neighbor 203.0.113.2 route-map FILTRE_ISP_OUT out

route-map FILTRE_ISP_OUT deny 10

match ip address 1

route-map FILTRE_ISP_OUT permit 20

```
R1(config-if)#router bgp 1
R1(config-router)#neighbor 203
*Mar  1 08:23:12.737: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2
in wrong AS) 2 bytes 11D7
R1(config-router)#neighbor 203 FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0
D7 00B4 C0A8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00
R1(config-router)#neighbor 203.0.113.2 route-map FILTRE_ISP_I
*Mar  1 08:23:39.257: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2
in wrong AS) 2 bytes 11D7
R1(config-router)#neighbor 203.0.113.2 route-map FILTRE_ISP_ISP FFFF FFFF FFFF
FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 C0A8 2E04 1002 0601 0400 0100 0102 0
02 0202 00
R1(config-router)#neighbor 203.0.113.2 route-map FILTRE_ISP_OUT out
R1(config-router)#route-map FILTRE_ISP_
*Mar  1 08:24:06.669: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2
in wrong AS) 2 bytes 11D7
R1(config-router)#route-map FILTRE_ISP_OUT FFFF FFFF FFFF FFFF FFFF FFFF FFFF
002D 0104 11D7 00B4 C0A8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00
R1(config-router)#route-map FILTRE_ISP_OUT deny 10
R1(config-route-map)#match ip address 1
R1(config-route-map)#route-map FILTRE
*Mar  1 08:24:40.225: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2
in wrong AS) 2 bytes 11D7
R1(config-route-map)#route-map FILTRE FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF
0104 11D7 00B4 C0A8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00
R1(config-route-map)#route-map FILTRE_ISP_OUT permit 20
R1(config-route-map)#EXIT
R1(config)#
*Mar  1 08:25:06.053: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2
R1(config)# FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 C0A8
R1(config)#
```

IV. Test et Validation

Verifions les ping et les traceroute

Ping de R1 VERS R4

```
R1#ping 192.168.14.4

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.14.4, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/12/16 ms
R1#
```

Ping de R1 VERS R2

```
R1#ping 192.168.12.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/15/32 ms
R1#
```

Ping de R1 VERS R3

```
2E04 1002 0001 0400 0100 0102 0200 0002 0202 00
R1#ping 192.168.13.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.13.3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/16/24 ms
R1#
```

Ping de R1 VERS ISP

```
1002 0002 0100 0100 0102 0100 0002 0100 00
R1#ping 203.0.113.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/21/24 ms
R1#
```

Ping de R2 VERS R1

```
R2#ping 192.168.12.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/19/24 ms
R2#
```

Ping de R2 VERS R3

```
R2#ping 192.168.23.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.23.3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/17/28 ms
R2#
```

Ping de R2 VERS R4

```
R2#ping 192.168.24.4

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.24.4, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/12/12 ms
R2#
```

Ping de R3 VERS R1

```
R3#ping 192.168.13.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.13.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/12/16 ms
R3#
```

Ping de R3 VERS R2

```
R3#ping 192.168.23.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.23.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/20/24 ms
R3#
```

Ping de R3 VERS R5

Projet technologie internet

p

```
R3#ping 192.168.35.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.35.5, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/9/12 ms
R3#
```

Ping de R4 VERS R1

```
R4#ping 192.168.14.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.14.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/20/32 ms
R4#
```

Ping de R4 VERS R2

```
R4#ping 192.168.24.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.24.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/10/12 ms
R4#
```

Ping de R4 VERS R5

```
R4#ping 192.168.45.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.45.5, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/12/20 ms
R4#
```

Ping de R5 VERS R3

```
R5#ping 192.168.35.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.35.3, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/17/24 ms
R5#
```

Ping de R5 VERS R4

```
R5#ping 192.168.45.4
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.45.4, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/23/32 ms
R5#
```

Ping de R5 VERS R7

```
R5#ping 192.168.57.7
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.57.7, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/11/16 ms
R5#
```

Ping de R6 VERS R7

```
R6#ping 192.168.67.7
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.67.7, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/21/24 ms
R6#
```

Ping de R7 VERS R6

```
R7#ping 192.168.67.6
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.67.6, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/11/16 ms
R7#
```

Ping de R7 VERS R5

```
R7#ping 192.168.57.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.57.5, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/15/32 ms
R7#
```

Ping AVEC Le serveur DNS

Ping de R4 vers L'AS 1 (via DNS)

```
R4#ping server.as1.sn

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.14.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/12/16 ms
R4#
```

Ping de R4 vers L'AS23 (via DNS)

```
R4#ping server.as23.sn

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.45.5, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/11/16 ms
R4#
```

Ping de R4 vers L'AS65505 (via DNS)

```
R4#ping server.as65505.sn

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.45.5, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/13/20 ms
R4#
```

OUFFF.....Enfin le ping marche pour tous les routeurs

À quoi sert Traceroute ?

Elle déterminer le **chemin emprunté par les paquets** pour atteindre une destination, en affichant chaque saut (routeur) traversé. Elle permet de :

- Vérifier la connectivité entre les AS (ex: AS1 → AS23 → AS4567 → AS65505).
- Identifier des **points de blocage** (ACLs mal configurées, routes manquantes).
- Valider les politiques de routage BGP et la redondance.

```
R2#sh ip bgp
BGP table version is 3, local router ID is 192.168.24.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop          Metric LocPrf Weight Path
*> 192.168.23.0      0.0.0.0              0         32768 ?
*> 203.0.113.0       192.168.12.1         0          0 1 ?
R2#
```

```
R3#sh ip bgp
BGP table version is 3, local router ID is 192.168.35.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop          Metric LocPrf Weight Path
*> 192.168.23.0      0.0.0.0              0         32768 ?
*> 203.0.113.0       192.168.13.1         0          0 1 ?
R3#
```

```
R5#sh ip bgp
BGP table version is 4, local router ID is 192.168.58.5
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop          Metric LocPrf Weight Path
*> 192.168.45.0      0.0.0.0              0         32768 ?
*> 192.168.46.0      192.168.45.4         1         32768 ?
*> 192.168.57.0      0.0.0.0              0         32768 ?
R5#
```

Simulation de panne (coupure de lien) et validation de la redondance du réseau

1. Choix du Lien à Couper

Pour couper un lien, on cherche un lien critique entre **R3 (AS23)** et **R4 (Sub-AS45 de l'AS4567)**.

On doit vérifier si le trafic se réoriente via un chemin alternatif (ex: R2 → R5 → R7 → AS65505).

2. Étapes de Simulation

Coupure du Lien

Sur R3 (AS23) : On désactive l'interface fastethernet1/0 relie a R4

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#int fa1/0
R3(config-if)#n
*Mar  1 10:53:32.953: %BGP-3-NOTIFICATION: sent to neighbor 192.168.35.5 2/2 (peer
wrong AS) 2 bytes 11D7
R3(config-if)#n FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00B4 C0A8
1002 0601 0400 0100 0102 0280 0002 0202 00
R3(config-if)#sh
R3(config-if)#
*Mar  1 10:53:39.533: %LINK-5-CHANGED: Interfa
```

Vérification de la Connectivité

Sur R1(AS1) on vérifie si on peut atteindre R5 via R3 : trace route 192.168.35.5

Projet technologie internet

p

```
R1#traceroute 192.168.35.5

Type escape sequence to abort.
Tracing the route to 192.168.35.5

 0  1
*Mar  1 11:12:49.421: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
in wrong AS) 2 bytes 11D7 FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00
B4 COA8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00 * * *
 2  * * *
 3  * * *
 4  *
*Mar  1 11:13:18.637: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
in wrong AS) 2 bytes 11D7 FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00
B4 COA8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00 * *
 5  * * *
 6  * * *
 7  * *
*Mar  1 11:13:49.017: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
in wrong AS) 2 bytes 11D7 FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00
B4 COA8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00 *
 8  * * *
 9  * * *
10  * *
*Mar  1 11:14:16.629: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
in wrong AS) 2 bytes 11D7 FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7 00
B4 COA8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00 *
11  * * *
12  * * *
13  * * *
14
*Mar  1 11:14:48.145: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
in wrong AS) 2 bytes 11D7 * FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7
00B4 COA8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00 * *
15  * * *
16  * * *
17  *
*Mar  1 11:15:17.877: %BGP-3-NOTIFICATION: sent to neighbor 192.168.14.4 2/2 (peer
in wrong AS) 2 bytes 11D7 * FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 11D7
00B4 COA8 2E04 1002 0601 0400 0100 0102 0280 0002 0202 00 *
18  * * *
19  * * *
```

En faisant le ping de R3 vers R5

```
R3#ping 192.168.35.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.35.5, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R3#
```

c. Analyse des Tables de Routage

Sur R3 (AS23) : Faisons **show ip route**

```
R3#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.13.0/24 is directly connected, FastEthernet0/0
B    203.0.113.0/24 [20/0] via 192.168.13.1, 10:57:57
     3.0.0.0/24 is subnetted, 1 subnets
C      3.3.3.0 is directly connected, Loopback0
C    192.168.23.0/24 is directly connected, FastEthernet0/1
R3#conf t
```

On voit que le chemin qui mène vers le Routeur R5 est absent

3. Validation de la Redondance

- **Temps de Convergence :**
 - BGP/OSPF/EIGRP devraient recalculer les routes en **quelques secondes**.
 - Vérifions avec show ip protocols pour confirmer les timers.

```
R3#sh ip protocol
Routing Protocol is "eigrp 23"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  EIGRP metric weight K1=1, K2=0, K3=1, K4=0, K5=0
  EIGRP maximum hopcount 100
  EIGRP maximum metric variance 1
  Redistributing: eigrp 23
  EIGRP NSF-aware route hold timer is 240s
  Automatic network summarization is in effect
  Maximum path: 4
  Routing for Networks:
    192.168.23.0
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: internal 90 external 170

Routing Protocol is "bgp 23"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  IGP synchronization is disabled
  Automatic route summarization is disabled
  Redistributing: eigrp 23
  Neighbor(s):
    Address          FiltIn FiltOut DistIn DistOut Weight RouteMap
    192.168.13.1      FiltIn FiltOut DistIn DistOut Weight RouteMap
    192.168.35.5      FiltIn FiltOut DistIn DistOut Weight RouteMap
  Maximum path: 1
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: external 20 internal 200 local 200

R3#
```

4. Rétablissement du Lien

- Sur R3 (AS23) : On reactive l'interface fa1/0

Vérification Post-Rétablissement

Tables de Routage : **show ip route**

Projet technologie internet

p

```
R3#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.13.0/24 is directly connected, FastEthernet0/0
B    203.0.113.0/24 [20/0] via 192.168.13.1, 10:58:45
     3.0.0.0/24 is subnetted, 1 subnets
C      3.3.3.0 is directly connected, Loopback0
C    192.168.23.0/24 is directly connected, FastEthernet0/1
C    192.168.35.0/24 is directly connected, FastEthernet1/0
R3#
```

On voit que la route que mène vers le Routeur R5 est rétablie

Test des Ping sur Wireshark entre R1 et R4

Capture en cours de Standard input [R1 FastEthernet1/1 to R4 FastEthernet0/0]

Fichier Editer Vue Aller Capture Analyser Statistiques Telephonie Wireless Outils Aide

Appliquer un filtre d'affichage ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	c4:01:1d:68:f1:01	c4:04:1f:b0:00:00	CDP	345	Device ID: R1 Port ID: FastEthernet1/1
2	0.424783	c4:04:1f:b0:00:00	c4:04:1f:b0:00:00	LOOP	60	Reply
3	10.284129	c4:04:1f:b0:00:00	c4:04:1f:b0:00:00	CDP	350	Device ID: R4 Port ID: FastEthernet0/0
4	10.423812	c4:04:1f:b0:00:00	c4:04:1f:b0:00:00	LOOP	60	Reply
5	17.760032	192.168.14.1	192.168.14.4	TCP	60	43445 → 179 [SYN] Seq=0 Win=16384 Len=0 MSS=1460
6	17.763821	192.168.14.4	192.168.14.1	TCP	60	179 → 43445 [SYN, ACK] Seq=0 Ack=1 Win=16384 Len=0 MSS=1460
7	17.770734	192.168.14.1	192.168.14.4	TCP	60	43445 → 179 [ACK] Seq=1 Ack=1 Win=16384 Len=0
8	17.783470	192.168.14.1	192.168.14.4	BGP	99	OPEN Message
9	17.787388	192.168.14.4	192.168.14.1	BGP	118	OPEN Message, KEEPALIVE Message
10	17.794165	192.168.14.1	192.168.14.4	BGP	77	NOTIFICATION Message
11	17.798001	192.168.14.4	192.168.14.1	TCP	60	179 → 43445 [FIN, PSH, ACK] Seq=65 Ack=69 Win=16316 Len=0
12	17.804828	192.168.14.1	192.168.14.4	TCP	60	43445 → 179 [ACK] Seq=69 Ack=66 Win=16320 Len=0
13	17.890762	192.168.14.1	192.168.14.4	TCP	60	43445 → 179 [FIN, PSH, ACK] Seq=69 Ack=66 Win=16320 Len=0
14	17.895689	192.168.14.4	192.168.14.1	TCP	60	179 → 43445 [ACK] Seq=66 Ack=70 Win=16316 Len=0
15	20.418511	c4:04:1f:b0:00:00	c4:04:1f:b0:00:00	LOOP	60	Reply

> Frame 1: 345 bytes on wire (2760 bits), 345 bytes captured (2760 bits) on interface -, id 0

> IEEE 802.3 Ethernet

> Logical-Link Control

> Cisco Discovery Protocol

0000 01 00 0c cc cc cc c4 01 1d 68 f1 01 01 4b aa aah...K..

0010 03 00 00 0c 20 00 02 b4 27 30 00 01 00 06 52 310....R1

0020 00 05 00 fb 43 69 73 63 6f 20 49 4f 53 20 53 6fCisco IOS So

0030 66 74 77 61 72 65 2c 20 33 37 30 30 20 53 6f 66 ftware, 3700 Sof

0040 74 77 61 72 65 20 28 43 33 37 34 35 2d 41 44 56 tware (C 3745-ADV

0050 45 4e 54 45 52 50 52 49 53 45 4b 39 2d 4d 29 2c ENTERPRI SEK9-M),

0060 20 56 65 72 73 69 6f 6e 20 31 32 2e 34 28 32 35 Version 12.4(25

0070 64 29 2c 20 52 45 4c 45 41 53 45 20 53 4f 46 54 d), RELE ASE SOFT

0080 57 41 52 45 20 28 66 63 31 29 0a 54 65 63 68 6e WARE (fc 1). Techn

0090 69 63 61 6c 20 53 75 70 70 6f 72 74 3a 20 68 74 ical Sup port: ht

00a0 74 70 3a 2f 2f 77 77 77 2e 63 69 73 63 6f 2e 63 tp://www .cisco.c

00b0 6f 6d 2f 74 65 63 68 73 75 70 70 6f 72 74 0a 43 om/techs upport-C

00c0 6f 70 79 72 69 67 68 74 20 28 63 29 20 31 39 38 opyright (c) 198

00d0 36 2d 32 30 31 30 20 62 79 20 43 69 73 63 6f 20 6-2010 b y Cisco

00e0 53 79 73 74 65 6d 73 2c 20 49 6e 63 2e 0a 43 6f Systems, Inc..Co

00f0 6d 70 69 6c 65 64 20 57 65 64 20 31 38 2d 41 75 mpiled w ed 18-Au

0100 67 2d 31 30 20 30 38 3a 31 38 20 62 79 20 72 72 g-10 08: 18 by pr

0110 6f 64 5f 72 65 6c 5f 74 65 61 6d 00 06 00 0e 43 od_rel_t eam...C

0120 69 73 63 6f 20 33 37 34 35 00 02 00 11 00 00 00 isco 374 5.....

0130 01 01 01 cc 00 04 c0 a8 0e 01 00 03 00 13 46 61Fa

Standard input: <live capture in progress>

Paquets : 15

Profil : Default

19°C

00:24

mardi:11/03/2025

Test des ping sur Wireshark entre R1 et R2

Projet technologie internet

p

Capture en cours de Standard input [R1 FastEthernet1/0 to R2 FastEthernet0/0]

Fichier Editer Vue Aller Capture Analyser Statistiques Telephonie Wireless Outils Aide

Appliquer un filtre d'affichage ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	c4:02:05:4c:00:00	c4:02:05:4c:00:00	LOOP	60	Reply
2	4.868151	192.168.12.2	192.168.12.1	BGP	73	KEEPALIVE Message
3	4.878757	192.168.12.1	192.168.12.2	BGP	73	KEEPALIVE Message
4	5.103572	192.168.12.2	192.168.12.1	TCP	60	179 → 24977 [ACK] Seq=20 Ack=20 Win=15035 Len=0
5	9.996680	c4:02:05:4c:00:00	c4:02:05:4c:00:00	LOOP	60	Reply
6	19.878520	c4:01:1d:68:f1:00	CDP/VTP/DTP/PagP/UD...	CDP	345	Device ID: R1 Port ID: FastEthernet1/0
7	20.008208	c4:02:05:4c:00:00	c4:02:05:4c:00:00	LOOP	60	Reply
8	30.007215	c4:02:05:4c:00:00	c4:02:05:4c:00:00	LOOP	60	Reply
9	40.018968	c4:02:05:4c:00:00	c4:02:05:4c:00:00	LOOP	60	Reply
10	49.834831	c4:02:05:4c:00:00	CDP/VTP/DTP/PagP/UD...	CDP	350	Device ID: R2 Port ID: FastEthernet0/0
11	50.007526	c4:02:05:4c:00:00	c4:02:05:4c:00:00	LOOP	60	Reply
12	59.999252	c4:02:05:4c:00:00	c4:02:05:4c:00:00	LOOP	60	Reply
13	64.871177	192.168.12.2	192.168.12.1	BGP	73	KEEPALIVE Message
14	64.879962	192.168.12.1	192.168.12.2	BGP	73	KEEPALIVE Message
15	65.075169	192.168.12.2	192.168.12.1	TCP	60	179 → 24977 [ACK] Seq=39 Ack=39 Win=15016 Len=0
16	70.001905	c4:02:05:4c:00:00	c4:02:05:4c:00:00	LOOP	60	Reply

> Frame 8: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
> Ethernet II, Src: c4:02:05:4c:00:00 (c4:02:05:4c:00:00), Dst: c4:02:05:4c:00:00 (c4:02:05:4c:00:00)
> Configuration Test Protocol (loopback)
> Data (40 bytes)

0000 c4 02 05 4c 00 00 c4 02 05 4c 00 00 90 00 00 00 ...L...L...
0010 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

Standard input: <live capture in progress> Paquets: 16 Profil: Default

Taper ici pour rechercher

Test des ping sur Wireshark entre R1 et R3

Capture en cours de Standard input [R1 FastEthernet0/1 to R3 FastEthernet0/0]

Fichier Editer Vue Aller Capture Analyser Statistiques Telephonie Wireless Outils Aide

Appliquer un filtre d'affichage ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	c4:01:22:b8:00:01	c4:01:22:b8:00:01	LOOP	60	Reply
2	0.479550	c4:03:27:a0:00:00	c4:03:27:a0:00:00	LOOP	60	Reply
3	9.981435	c4:01:22:b8:00:01	c4:01:22:b8:00:01	LOOP	60	Reply
4	10.479533	c4:03:27:a0:00:00	c4:03:27:a0:00:00	LOOP	60	Reply
5	14.616776	192.168.13.1	192.168.13.3	BGP	73	KEEPALIVE Message
6	14.623604	192.168.13.3	192.168.13.1	BGP	73	KEEPALIVE Message
7	14.830893	192.168.13.1	192.168.13.3	TCP	60	43813 → 179 [ACK] Seq=20 Ack=20 Win=15187 Len=0
8	19.974031	c4:01:22:b8:00:01	c4:01:22:b8:00:01	LOOP	60	Reply
9	20.471845	c4:03:27:a0:00:00	c4:03:27:a0:00:00	LOOP	60	Reply
10	24.746805	c4:01:22:b8:00:01	CDP/VTP/DTP/PagP/UD...	CDP	350	Device ID: R1 Port ID: FastEthernet0/1
11	29.978677	c4:01:22:b8:00:01	c4:01:22:b8:00:01	LOOP	60	Reply
12	30.475797	c4:03:27:a0:00:00	c4:03:27:a0:00:00	LOOP	60	Reply
13	34.694155	c4:03:27:a0:00:00	CDP/VTP/DTP/PagP/UD...	CDP	350	Device ID: R3 Port ID: FastEthernet0/0
14	39.990343	c4:01:22:b8:00:01	c4:01:22:b8:00:01	LOOP	60	Reply
15	40.479289	c4:03:27:a0:00:00	c4:03:27:a0:00:00	LOOP	60	Reply
16	42.454534	c4:03:27:a0:00:00	DEC-NOP-Remote-Cons...	0x6002	77	DEC DNA Remote Console

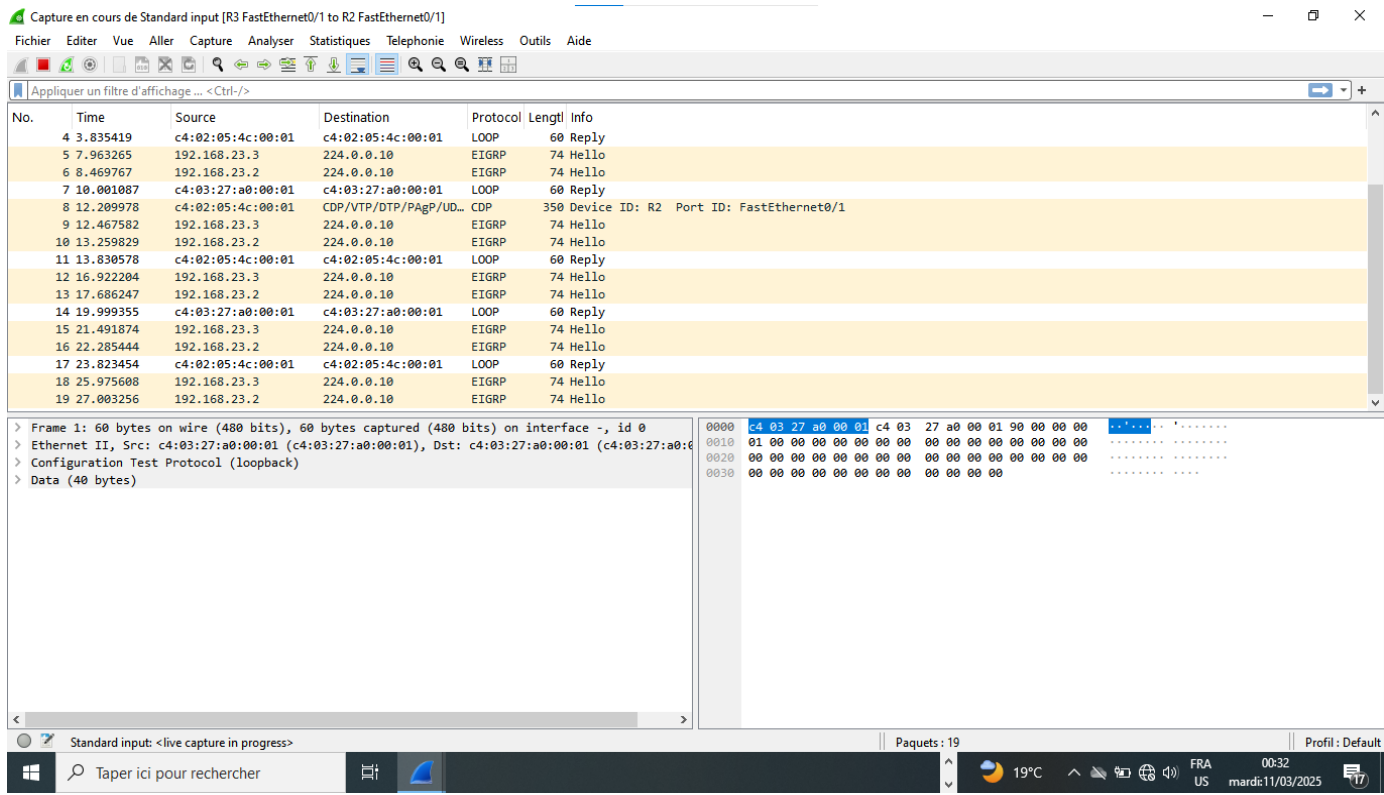
> Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
> Ethernet II, Src: c4:01:22:b8:00:01 (c4:01:22:b8:00:01), Dst: c4:01:22:b8:00:01 (c4:01:22:b8:00:01)
> Configuration Test Protocol (loopback)
> Data (40 bytes)

0000 c4 01 22 b8 00 01 c4 01 22 b8 00 01 00 00 00 00
0010 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

Standard input: <live capture in progress> Paquets: 16 Profil: Default

Taper ici pour rechercher

Test des ping sur Wireshark entre R2 et R4



Projet technologie internet

p

Capture en cours de Standard input [R2 FastEthernet1/0 to R4 FastEthernet1/1]

Fichier Editor Vue Aller Capture Analyser Statistiques Telephonie Wireless Outils Aide

Appliquer un filtre d'affichage ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
5	0.146477	192.168.24.2	192.168.24.4	TCP	60	179 → 49020 [FIN, PSH, ACK] Seq=24 Ack=47 Win=16339 Len=0
6	0.156242	192.168.24.4	192.168.24.2	TCP	60	49020 → 179 [ACK] Seq=47 Ack=25 Win=16361 Len=0
7	10.464878	c4:02:24:20:f1:00	CDP/VTP/DTP/PagP/UD...	CDP	345	Device ID: R2 Port ID: FastEthernet1/0
8	11.097030	c4:04:26:bc:f1:01	CDP/VTP/DTP/PagP/UD...	CDP	345	Device ID: R4 Port ID: FastEthernet1/1
9	32.436458	192.168.24.4	192.168.24.2	TCP	60	57988 → 179 [SYN] Seq=0 Win=16384 Len=0 MSS=1460
10	32.438293	192.168.24.2	192.168.24.4	TCP	60	45842 → 179 [SYN] Seq=0 Win=16384 Len=0 MSS=1460
11	32.446867	192.168.24.4	192.168.24.2	TCP	60	179 → 45842 [SYN, ACK] Seq=0 Ack=1 Win=16384 Len=0 MSS=1460
12	32.448700	192.168.24.2	192.168.24.4	TCP	60	179 → 57988 [SYN, ACK] Seq=0 Ack=1 Win=16384 Len=0 MSS=1460
13	32.457615	192.168.24.4	192.168.24.2	TCP	60	57988 → 179 [ACK] Seq=1 Ack=1 Win=16384 Len=0
14	32.459462	192.168.24.2	192.168.24.4	TCP	60	45842 → 179 [ACK] Seq=1 Ack=1 Win=16384 Len=0
15	32.468356	192.168.24.4	192.168.24.2	BGP	99	OPEN Message
16	32.470183	192.168.24.2	192.168.24.4	BGP	99	OPEN Message
17	32.479098	192.168.24.4	192.168.24.2	TCP	60	179 → 45842 [RST] Seq=1 Win=16384 Len=0
18	32.480946	192.168.24.2	192.168.24.4	TCP	60	179 → 57988 [RST] Seq=1 Win=16384 Len=0
19	32.489835	192.168.24.4	192.168.24.2	TCP	60	179 → 45842 [RST] Seq=1 Win=0 Len=0
20	32.491688	192.168.24.2	192.168.24.4	TCP	60	179 → 57988 [RST] Seq=1 Win=0 Len=0

> Frame 1: 99 bytes on wire (792 bits), 99 bytes captured (792 bits) on interface -, id 0

> Ethernet II, Src: c4:04:26:bc:f1:01 (c4:04:26:bc:f1:01), Dst: c4:02:24:20:f1:00 (c4:02:24:20:f1:00)

> Internet Protocol Version 4, Src: 192.168.24.4, Dst: 192.168.24.2

> Transmission Control Protocol, Src Port: 49020, Dst Port: 179, Seq: 1, Ack: 1, Len: 45

> Border Gateway Protocol - OPEN Message

Standard input: <live capture in progress> Paquets: 20 Profil: Default

Taper ici pour rechercher

19°C 00:34 FRA US mardi 11/03/2025

Test des ping sur Wireshark entre R3 et R5

Projet technologie internet

p

Capture en cours de Standard input [R3 FastEthernet1/0 to R5 FastEthernet0/0]

Fichier Editer Vue Aller Capture Analyser Statistiques Telephonie Wireless Outils Aide

Appliquer un filtre d'affichage ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
13	25.424170	192.168.35.5	192.168.35.3	TCP	60	30016 → 179 [ACK] Seq=47 Ack=25 Win=16361 Len=0
14	30.923057	c4:05:1a:08:00:00	CDP/VTP/DTP/PagP/UD...	CDP	350	Device ID: R5 Port ID: FastEthernet0/0
15	31.030392	c4:05:1a:08:00:00	c4:05:1a:08:00:00	LOOP	60	Reply
16	41.052247	c4:05:1a:08:00:00	c4:05:1a:08:00:00	LOOP	60	Reply
17	51.006320	c4:05:1a:08:00:00	c4:05:1a:08:00:00	LOOP	60	Reply
18	52.484414	192.168.35.5	192.168.35.3	TCP	60	41957 → 179 [SYN] Seq=0 Win=16384 Len=0 MSS=1460
19	52.496130	192.168.35.3	192.168.35.5	TCP	60	179 → 41957 [SYN, ACK] Seq=0 Ack=1 Win=16384 Len=0 MSS=1460
20	52.506873	192.168.35.5	192.168.35.3	TCP	60	41957 → 179 [ACK] Seq=1 Ack=1 Win=16384 Len=0
21	52.517615	192.168.35.5	192.168.35.3	BGP	99	OPEN Message
22	52.530328	192.168.35.3	192.168.35.5	BGP	77	NOTIFICATION Message
23	52.541050	192.168.35.5	192.168.35.3	TCP	60	41957 → 179 [FIN, PSH, ACK] Seq=46 Ack=24 Win=16361 Len=0
24	52.551796	192.168.35.3	192.168.35.5	TCP	60	179 → 41957 [ACK] Seq=24 Ack=47 Win=16339 Len=0
25	52.626984	192.168.35.3	192.168.35.5	TCP	60	179 → 41957 [FIN, PSH, ACK] Seq=24 Ack=47 Win=16339 Len=0
26	52.637721	192.168.35.5	192.168.35.3	TCP	60	41957 → 179 [ACK] Seq=47 Ack=25 Win=16361 Len=0
27	60.005831	c4:03:0c:7c:f1:00	CDP/VTP/DTP/PagP/UD...	CDP	350	Device ID: R3 Port ID: FastEthernet1/0
28	61.023897	c4:05:1a:08:00:00	c4:05:1a:08:00:00	LOOP	60	Reply

> Frame 1: 350 bytes on wire (2800 bits), 350 bytes captured (2800 bits) on interface -, id 0
> IEEE 802.3 Ethernet
> Logical-Link Control
> Cisco Discovery Protocol

0000 01 00 0c cc cc cc c4 03 0c 7c f1 00 01 50 aa aa |...P...
0010 03 00 00 0c 20 00 02 b4 ff 2e 00 01 00 06 52 33R3
0020 00 05 00 fb 43 69 73 63 6f 20 49 4f 53 20 53 6fCisco IOS So
0030 66 74 77 61 72 65 2c 20 33 37 30 30 20 53 6f 66 ftware, 3700 Sof
0040 74 77 61 72 65 20 28 43 33 37 34 35 2d 41 44 56 tware (C 3745-ADV
0050 45 4e 54 45 52 50 52 49 53 45 4b 39 2d 4d 29 2c ENTERPRI SEK9-HV),
0060 20 56 65 72 73 69 6f 6e 20 31 32 2e 34 28 32 35 Version 12.4(25
0070 64 29 2c 20 52 45 4c 45 41 53 45 20 53 4f 46 54 d), RELE ASE SOFT
0080 57 41 52 45 20 28 66 63 31 29 0a 54 65 63 68 6e WARE (fc 1). Techn
0090 69 63 61 6c 20 53 75 70 70 6f 72 74 3a 20 60 74 ical Sup ports: ht
00a0 74 70 3a 2f 2f 77 77 77 2e 63 69 73 63 6f 2e 63 tpi://www .cisco.c
00b0 6f 6d 2f 74 65 63 68 73 75 70 70 6f 72 74 0a 43 om/techs upport:C
00c0 6f 70 79 72 69 6f 68 74 20 28 63 29 20 31 39 38 opyright (c) 198
00d0 36 2d 32 30 31 30 20 62 79 20 43 69 73 63 6f 20 6-2010 b y Cisco
00e0 53 79 73 74 65 6d 73 2c 20 49 6e 63 2e 0a 43 6f Systems, Inc..Co
00f0 6d 70 69 6c 65 64 20 57 65 64 20 31 38 2d 41 75 mpiled Wed 18-Au
0100 67 2d 31 30 20 30 38 3a 31 38 20 62 79 20 70 72 g-10 08: 18 by pr
0110 6f 64 5f 72 65 6c 5f 74 65 61 6d 00 06 00 0e 43 od_rel_t eam....C
0120 69 73 63 6f 20 33 37 34 35 00 02 00 11 00 00 00 isco 374 5.....C
0130 01 01 01 cc 00 04 c0 a8 23 03 00 03 00 13 46 61#.....Fa

Standard input: <live capture in progress> Paquets: 28 Profil: Default

Taper ici pour rechercher 19°C FRA US 00:37 mardi:11/03/2025

Test des ping sur Wireshark entre R4 et R5

Projet technologie internet

p

Capture en cours de Standard input [R5 FastEthernet0/1 to R4 FastEthernet0/1]

Fichier Editor Vue Aller Capture Analyser Statistiques Telephonie Wireless Outils Aide

Appliquer un filtre d'affichage ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
4	11.282053	c4:04:1f:b0:00:01	c4:04:1f:b0:00:01	LOOP	60	Reply
5	16.442235	c4:05:1a:08:00:01	c4:05:1a:08:00:01	LOOP	60	Reply
6	21.189192	c4:04:1f:b0:00:01	c4:04:1f:b0:00:01	CDP/VTP/DTP/PAGP/UD...	350	Device ID: R4 Port ID: FastEthernet0/1
7	21.264387	c4:04:1f:b0:00:01	c4:04:1f:b0:00:01	LOOP	60	Reply
8	21.607801	192.168.45.5	224.0.0.9	RIPv2	66	Response
9	26.432815	c4:05:1a:08:00:01	c4:05:1a:08:00:01	LOOP	60	Reply
10	28.278492	192.168.45.4	224.0.0.9	RIPv2	66	Response
11	31.292164	c4:04:1f:b0:00:01	c4:04:1f:b0:00:01	LOOP	60	Reply
12	36.429816	c4:05:1a:08:00:01	c4:05:1a:08:00:01	LOOP	60	Reply
13	41.303631	c4:04:1f:b0:00:01	c4:04:1f:b0:00:01	LOOP	60	Reply
14	46.327354	c4:05:1a:08:00:01	c4:05:1a:08:00:01	CDP/VTP/DTP/PAGP/UD...	350	Device ID: R5 Port ID: FastEthernet0/1
15	46.434860	c4:05:1a:08:00:01	c4:05:1a:08:00:01	LOOP	60	Reply
16	49.556634	192.168.45.5	224.0.0.9	RIPv2	66	Response
17	51.276053	c4:04:1f:b0:00:01	c4:04:1f:b0:00:01	LOOP	60	Reply
18	56.437529	c4:05:1a:08:00:01	c4:05:1a:08:00:01	LOOP	60	Reply
19	56.634208	192.168.45.4	224.0.0.9	RIPv2	66	Response

> Frame 1: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface -, id 0

> Ethernet II, Src: c4:04:1f:b0:00:01 (c4:04:1f:b0:00:01), Dst: IPv4mcast_09 (01:00:5e:00:00:09)

> Internet Protocol Version 4, Src: 192.168.45.4, Dst: 224.0.0.9

> User Datagram Protocol, Src Port: 520, Dst Port: 520

> Routing Information Protocol

0000 01 00 5e 00 00 09 c4 04 1f b0 00 01 08 00 45 c0 ..^.....E..

0010 00 34 00 00 00 02 11 ea 43 e0 a8 2d 04 e0 00 ..4.....C...

0020 00 09 02 08 02 08 00 20 3e 39 02 02 00 00 02>9.....

0030 00 00 c0 a8 2e 00 ff ff ff 00 00 00 00 00 00>.....

0040 00 01

Standard input: <live capture in progress>

Paquets : 19

Profil: Default

19°C

FRA US

00:39

mardi:11/03/2025

Test des ping sur Wireshark entre R4 et R6

Projet technologie internet

p

Capture en cours de Standard input [R4 FastEthernet1/0 to R6 FastEthernet0/0]

Fichier Editer Vue Aller Capture Analyser Statistiques Telephonie Wireless Outils Aide

Appliquer un filtre d'affichage... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	c4:04:26:bc:f1:00	c4:04:26:bc:f1:00	CDP	345	Device ID: R4 Port ID: FastEthernet1/0
2	3.404949	192.168.46.4	224.0.0.9	RIPv2	86	Response
3	5.082345	c4:06:1c:5c:00:00	c4:06:1c:5c:00:00	LOOP	60	Reply
4	15.064745	c4:06:1c:5c:00:00	c4:06:1c:5c:00:00	LOOP	60	Reply
5	25.091568	c4:06:1c:5c:00:00	c4:06:1c:5c:00:00	LOOP	60	Reply
6	32.609837	192.168.46.4	224.0.0.9	RIPv2	86	Response
7	35.095677	c4:06:1c:5c:00:00	c4:06:1c:5c:00:00	LOOP	60	Reply
8	44.674090	c4:06:1c:5c:00:00	c4:06:1c:5c:00:00	CDP	350	Device ID: R6 Port ID: FastEthernet0/0
9	45.082277	c4:06:1c:5c:00:00	c4:06:1c:5c:00:00	LOOP	60	Reply
10	55.071279	c4:06:1c:5c:00:00	c4:06:1c:5c:00:00	LOOP	60	Reply
11	58.797980	192.168.46.4	224.0.0.9	RIPv2	86	Response
12	60.021429	c4:04:26:bc:f1:00	c4:04:26:bc:f1:00	CDP	345	Device ID: R4 Port ID: FastEthernet1/0
13	65.060972	c4:06:1c:5c:00:00	c4:06:1c:5c:00:00	LOOP	60	Reply
14	75.093610	c4:06:1c:5c:00:00	c4:06:1c:5c:00:00	LOOP	60	Reply

> Frame 1: 345 bytes on wire (2760 bits), 345 bytes captured (2760 bits) on interface -, id 0
> IEEE 802.3 Ethernet
> Logical-Link Control
> Cisco Discovery Protocol

0000 01 00 0c cc cc cc c4 04 26 bc f1 00 01 4b aa aa ... &...K...
0010 03 00 00 0c 20 00 02 b4 08 2a 00 01 00 06 52 34 ... *...R4
0020 00 05 00 fb 43 69 73 63 6f 20 49 4f 53 20 53 6f ...Cisc o IOS So
0030 66 74 77 61 72 65 2c 20 33 37 30 30 20 53 6f 66 ...ftware, 3700 Sof
0040 74 77 61 72 65 20 28 43 33 37 34 35 2d 41 44 56 ...tware (C 3745-ADV
0050 45 4e 54 45 52 50 52 49 53 45 4b 39 2d 4d 29 2c ...ENTERPRI SEK9-M),
0060 20 56 65 72 73 69 6f 6e 20 31 32 2e 34 28 32 35 ...Version 12.4(25
0070 64 29 2c 20 52 45 4c 45 41 53 45 20 53 4f 46 54 ...d), RELE ASE SOFT
0080 57 41 52 45 20 28 66 63 31 29 0a 54 65 63 68 6e ...WARE (fc l) Techn
0090 69 63 61 6c 20 53 75 70 70 6f 72 74 3a 20 68 74 ...ical Sup port: ht
00a0 74 70 3a 2f 2f 77 77 2e 63 69 73 63 6f 2e 63 ...tp://www .cisco.c
00b0 6f 6d 2f 74 65 63 68 73 75 70 70 6f 72 74 0a 43 ...om/techs upport:C
00c0 6f 70 79 72 69 67 68 74 20 28 63 29 20 31 39 38 ...opyright (c) 198
00d0 36 2d 32 30 31 30 20 62 79 20 43 69 73 63 6f 20 ...6-2010 b y Cisco
00e0 53 79 73 74 65 6d 73 2c 20 49 6e 63 2e 0a 43 6f ...Systems, Inc. Co
00f0 6d 70 69 6c 65 64 20 57 65 64 20 31 38 2d 41 75 ...mpiled W ed 18-Au
0100 67 2d 31 30 20 30 38 3a 31 38 20 62 79 20 70 72 ...g-10 08: 18 by pr
0110 6f 64 5f 72 65 6c 5f 74 65 61 6d 00 06 00 0e 43 ...od_rel_t eam...C
0120 69 73 63 6f 20 33 37 34 35 00 02 00 11 00 00 00 ...isco 374 5...
0130 01 01 01 cc 00 04 c0 a8 2e 04 00 03 00 13 46 61Fa

Standard input: <live capture in progress> Paquets: 14 Profil: Default

Taper ici pour rechercher

19°C 00:42 FRA US mardi 11/03/2025

Test des ping sur Wireshark entre R6 et R7

Projet technologie internet

p

Capture en cours de Standard input [R6 FastEthernet0/1 to R7 FastEthernet0/0]

Fichier Editer Vue Aller Capture Analyser Statistiques Telephonie Wireless Outils Aide

Appliquer un filtre d'affichage ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.67.6	224.0.0.5	OSPF	94	Hello Packet
2	1.675392	c4:06:1c:5c:00:01	c4:06:1c:5c:00:01	CDP	350	Device ID: R6 Port ID: FastEthernet0/1
3	2.089517	c4:06:1c:5c:00:01	c4:06:1c:5c:00:01	LOOP	60	Reply
4	5.350547	c4:07:2a:c8:00:00	c4:07:2a:c8:00:00	LOOP	60	Reply
5	7.680240	192.168.67.7	224.0.0.5	OSPF	94	Hello Packet
6	9.995186	192.168.67.6	224.0.0.5	OSPF	94	Hello Packet
7	12.100335	c4:06:1c:5c:00:01	c4:06:1c:5c:00:01	LOOP	60	Reply
8	15.351604	c4:07:2a:c8:00:00	c4:07:2a:c8:00:00	LOOP	60	Reply
9	17.692552	192.168.67.7	224.0.0.5	OSPF	94	Hello Packet
10	19.978291	192.168.67.6	224.0.0.5	OSPF	94	Hello Packet
11	22.083033	c4:06:1c:5c:00:01	c4:06:1c:5c:00:01	LOOP	60	Reply
12	25.339125	c4:07:2a:c8:00:00	c4:07:2a:c8:00:00	LOOP	60	Reply
13	27.685705	192.168.67.7	224.0.0.5	OSPF	94	Hello Packet
14	30.000112	192.168.67.6	224.0.0.5	OSPF	94	Hello Packet
15	32.077556	c4:06:1c:5c:00:01	c4:06:1c:5c:00:01	LOOP	60	Reply
16	35.352481	c4:07:2a:c8:00:00	c4:07:2a:c8:00:00	LOOP	60	Reply

> Frame 1: 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface -, id 0
> Ethernet II, Src: c4:06:1c:5c:00:01 (c4:06:1c:5c:00:01), Dst: IPv4mcast_05 (01:00:5e:00:00:05)
> Internet Protocol Version 4, Src: 192.168.67.6, Dst: 224.0.0.5
> Open Shortest Path First

0000 01 00 5e 00 00 05 c4 06 1c 5c 00 01 08 00 45 c0 ..A.....E..
0010 00 50 0f b9 00 00 01 59 c5 28 c0 a8 43 06 e0 00 .P.....Y.(.C..
0020 00 05 02 01 00 30 c0 a8 2d 06 00 00 00 00 fd dc0:.....
0030 00 00 00 00 00 00 00 00 00 ff ff ff 00 00 0aC...C..
0040 12 01 00 00 28 c0 a8 43 06 c0 a8 43 07 c0 a89.....
0050 39 07 ff f6 00 03 00 01 00 04 00 00 00 01

Standard input: <live capture in progress> Paquets: 16 Profil: Default

Taper ici pour rechercher

19°C FRA US 00:44 mardi:11/03/2025

Test des ping sur Wireshark entre R5 et R7

Projet technologie internet

p

Capture en cours de Standard input [R7 FastEthernet0/1 to R5 FastEthernet1/0]

Fichier Editor Vue Aller Capture Analyser Statistiques Telephonie Wireless Outils Aide

Appliquer un filtre d'affichage ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.57.7	224.0.0.5	OSPF	90	Hello Packet
2	7.845546	c4:07:2a:c8:00:01	c4:07:2a:c8:00:01	LOOP	60	Reply
3	10.001729	192.168.57.7	224.0.0.5	OSPF	90	Hello Packet
4	10.355968	192.168.57.5	224.0.0.9	RIPv2	86	Response
5	16.344036	192.168.57.7	192.168.57.5	TCP	60	11491 → 179 [SYN] Seq=0 Win=16384 Len=0 MSS=1460
6	16.354904	192.168.57.5	192.168.57.7	TCP	60	179 → 11491 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
7	17.878140	c4:07:2a:c8:00:01	c4:07:2a:c8:00:01	LOOP	60	Reply
8	19.989045	192.168.57.7	224.0.0.5	OSPF	90	Hello Packet
9	23.279893	c4:07:2a:c8:00:01	c4:07:2a:c8:00:01	CDP/VTP/DTP/PagP/UD...	350	Device ID: R7 Port ID: FastEthernet0/1
10	27.840791	c4:07:2a:c8:00:01	c4:07:2a:c8:00:01	LOOP	60	Reply
11	29.985372	192.168.57.7	224.0.0.5	OSPF	90	Hello Packet
12	37.873154	c4:07:2a:c8:00:01	c4:07:2a:c8:00:01	LOOP	60	Reply
13	39.997077	192.168.57.7	224.0.0.5	OSPF	90	Hello Packet
14	40.287309	192.168.57.5	224.0.0.9	RIPv2	86	Response
15	42.201235	192.168.57.7	192.168.57.5	TCP	60	37155 → 179 [SYN] Seq=0 Win=16384 Len=0 MSS=1460
16	42.203105	192.168.57.5	192.168.57.7	TCP	60	179 → 37155 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

> Frame 1: 90 bytes on wire (720 bits), 90 bytes captured (720 bits) on interface -, id 0
> Ethernet II, Src: c4:07:2a:c8:00:01 (c4:07:2a:c8:00:01), Dst: IPv4mcast_05 (01:00:5e:00:00:05)
> Internet Protocol Version 4, Src: 192.168.57.7, Dst: 224.0.0.5
> Open Shortest Path First

0000 01 00 5e 00 00 05 c4 07 2a c8 00 01 08 00 45 c0 ...^.....*.....E.
0010 00 4c 23 bb 00 00 01 59 bb 29 c0 a8 39 07 60 00 ...L#...Y...)..9..
0020 00 05 02 01 00 2c c0 a8 39 07 00 00 00 00 f9 3e ...:.....9.....>
0030 00 00 00 00 00 00 00 00 00 00 ff ff 00 00 0a(..9.....
0040 12 01 00 00 00 28 c0 a8 39 07 00 00 00 00 ff f6
0050 00 03 00 01 00 04 00 00 00 01>

Standard input: <live capture in progress> Paquets: 16 Profil: Default

Taper ici pour rechercher

19°C FRA US 00:45 mardi 11/03/2025

Test des ping sur Wireshark entre R5 et R8

Projet technologie internet

p

Capture en cours de Standard input [R5 FastEthernet1/1 to R8 FastEthernet0/0]

Fichier Editer Vue Aller Capture Analyser Statistiques Telephonie Wireless Outils Aide

Appliquer un filtre d'affichage ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	c4:05:1d:10:f1:01	Broadcast	ARP	60	Who has 192.168.58.3? Tell 192.168.58.5
2	2.022424	c4:05:1d:10:f1:01	Broadcast	ARP	60	Who has 192.168.58.3? Tell 192.168.58.5
3	6.503361	c4:08:24:90:00:00	c4:08:24:90:00:00	LOOP	60	Reply
4	16.798637	c4:08:24:90:00:00	c4:08:24:90:00:00	LOOP	60	Reply
5	27.087654	c4:08:24:90:00:00	c4:08:24:90:00:00	LOOP	60	Reply
6	29.579014	c4:08:24:90:00:00	c4:08:24:90:00:00	LOOP	60	Reply
7	31.704993	c4:05:1d:10:f1:01	Broadcast	ARP	60	Who has 192.168.58.3? Tell 192.168.58.5
8	35.033556	c4:05:1d:10:f1:01	Broadcast	ARP	60	Who has 192.168.58.3? Tell 192.168.58.5
9	37.030980	c4:05:1d:10:f1:01	Broadcast	ARP	60	Who has 192.168.58.3? Tell 192.168.58.5
10	37.250704	c4:08:24:90:00:00	c4:08:24:90:00:00	LOOP	60	Reply
11	47.495583	c4:08:24:90:00:00	c4:08:24:90:00:00	LOOP	60	Reply
12	57.524769	c4:08:24:90:00:00	c4:08:24:90:00:00	LOOP	60	Reply
13	67.352203	c4:05:1d:10:f1:01	Broadcast	ARP	60	Who has 192.168.58.3? Tell 192.168.58.5
14	67.820760	c4:08:24:90:00:00	c4:08:24:90:00:00	LOOP	60	Reply
15	69.361821	c4:05:1d:10:f1:01	Broadcast	ARP	60	Who has 192.168.58.3? Tell 192.168.58.5
16	78.126582	c4:08:24:90:00:00	c4:08:24:90:00:00	LOOP	60	Reply

> Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
> Ethernet II, Src: c4:05:1d:10:f1:01 (c4:05:1d:10:f1:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Address Resolution Protocol (request)

```
0000  ff ff ff ff ff c4 05 1d 10 f1 01 08 06 00 01 .....
0010  08 00 06 04 00 01 c4 05 1d 10 f1 01 c0 a8 3a 05 .....
0020  00 00 00 00 00 00 c0 a8 3a 03 00 00 00 00 00 00 .....
0030  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
```

Standard input: <live capture in progress> Paquets: 16 Profil: Default

Taper ici pour rechercher

19°C 00:48 FRA US mardi 11/03/2025

Test des ping sur Wireshark entre R1 et ISP

Projet technologie internet

p

Capture en cours de Standard input [R1 FastEthernet0/0 to ISP FastEthernet0/0]

Fichier Editer Vue Aller Capture Analyser Statistiques Telephonie Wireless Outils Aide

Appliquer un filtre d'affichage ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	c2:0a:15:14:00:00	c2:0a:15:14:00:00	LOOP	60	Reply
2	0.250468	203.0.113.1	224.0.0.5	OSPF	90	Hello Packet
3	3.809961	c4:01:22:b8:00:00	c4:01:22:b8:00:00	LOOP	60	Reply
4	9.463422	203.0.113.1	203.0.113.2	BGP	73	KEEPALIVE Message
5	9.475145	203.0.113.2	203.0.113.1	BGP	73	KEEPALIVE Message
6	9.710403	203.0.113.1	203.0.113.2	TCP	60	50776 → 179 [ACK] Seq=20 Ack=20 Win=16061 Len=0
7	10.002367	c2:0a:15:14:00:00	c2:0a:15:14:00:00	LOOP	60	Reply
8	10.247227	203.0.113.1	224.0.0.5	OSPF	90	Hello Packet
9	13.803213	c4:01:22:b8:00:00	c4:01:22:b8:00:00	LOOP	60	Reply
10	14.715007	c2:0a:15:14:00:00	c2:0a:15:14:00:00	CDP	353	Device ID: ISP Port ID: FastEthernet0/0
11	17.523183	c4:01:22:b8:00:00	c4:01:22:b8:00:00	CDP	350	Device ID: R1 Port ID: FastEthernet0/0
12	19.985767	c2:0a:15:14:00:00	c2:0a:15:14:00:00	LOOP	60	Reply
13	20.244019	203.0.113.1	224.0.0.5	OSPF	90	Hello Packet
14	22.027784	c4:01:22:b8:00:00	c4:01:22:b8:00:00	DEC-NOP-Remote-Cons...	77	DEC DNA Remote Console
15	23.789310	c4:01:22:b8:00:00	c4:01:22:b8:00:00	LOOP	60	Reply

> Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
> Ethernet II, Src: c2:0a:15:14:00:00 (c2:0a:15:14:00:00), Dst: c2:0a:15:14:00:00 (c2:0a:15:14:00:00)
> Configuration Test Protocol (loopback)
> Data (40 bytes)

Standard input: <live capture in progress>

Paquets: 15

Profil: Default

19°C

FRA US

00:50

mardi 11/03/2025

Test des ping sur Wireshark entre R9 et R8

P



En résumé, on peut dire que ce projet a été enrichissant ça englobe tous ce qu'on a vu durant ce semestre on peut dire même que c'est la synthèse de tous les travaux pratiques qu'on a eu à faire. Nous avons entrepris la conception et la configuration d'un réseau d'entreprise multi-AS, en utilisant des protocoles de routage avancés pour optimiser la communication inter-AS et intra-AS. Nous avons réussi à mettre en place une topologie réseau répondant aux exigences spécifiques de chaque AS, assurant ainsi une connectivité efficace et la disponibilité des services critiques. Les tests de connectivité entre les différents pc de différents AS et les traces ont confirmé la connectivité de tous les As de l'entreprise. Pour l'amélioration future, nous recommandons l'intégration de technologies de sécurité avancées, le renforcement des protocoles de routage pour une convergence encore plus rapide, et une exploration plus poussée des services basés sur le cloud pour une flexibilité et une scalabilité accrues. Ces actions garantiront la pérennité et l'efficacité de notre infrastructure réseau dans un environnement technologique en constante évolution.